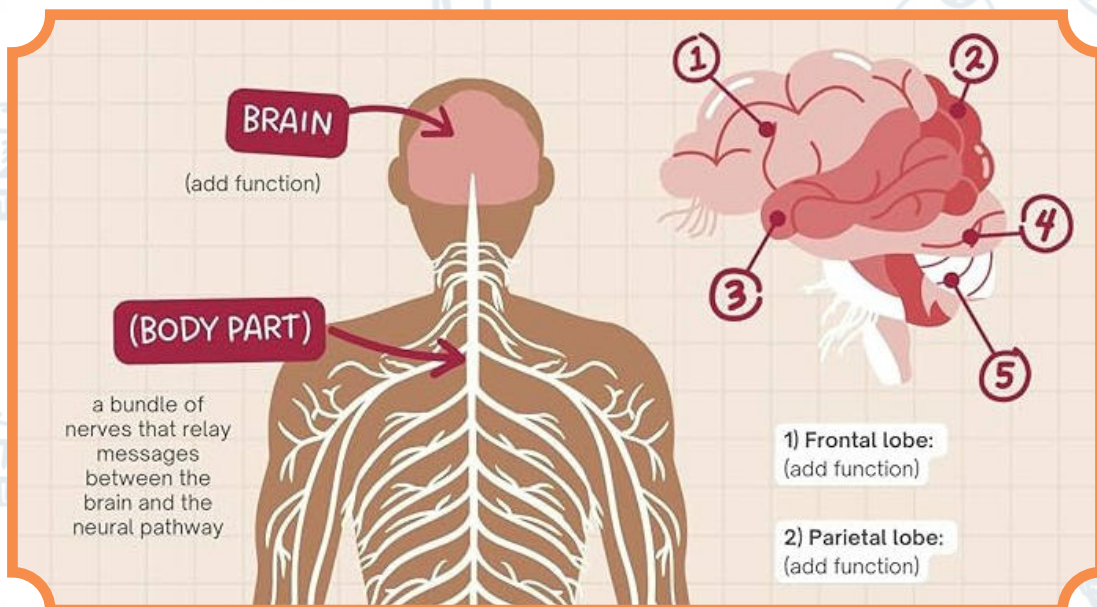


UNIT

17

NERVOUS COORDINATION





All animals except sponges have a network of nerve cells (nervous system) that:

- Gather information about body conditions and the external environment.
- Process and integrate that information.
- Send commands to muscles and glands for appropriate responses.

Steps Involved in Nervous Coordination

Definition:

The **nervous system** is the body's system for communication and coordination, using **electrical and chemical signals**.

Key Component:

- The system is made up of **highly specialized** cells called **neurons**.

Neurons (nerve cells) – fundamental units of the nervous system.

Each neuron performs **five essential functions**:

1. Receive Information

- From the **internal environment** (inside the body),
- From the **external environment** (outside the body), or
- From **other neurons**.

2. Integrate Information

- Process the incoming signals.
- Decide on an **appropriate output signal**.

3. Conduct the Signal

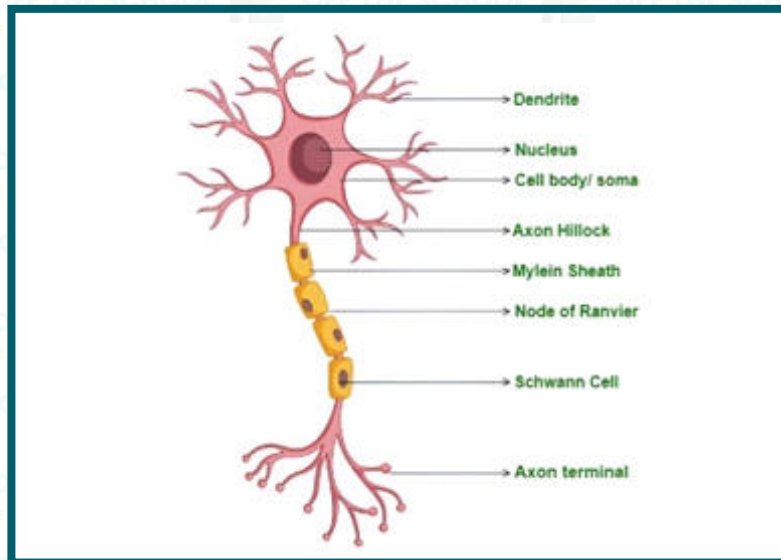
- Carry the signal along the length of the neuron
- To its **terminal ending** (may be some distance away).

4. Transmit the Signal

- Pass the signal to another **nerve cell, gland, or muscle**.

5. Coordinate Metabolic Activities

- Manage the internal **cellular activities** that keep the neuron functional and healthy



Receptors as Transducers

Definition

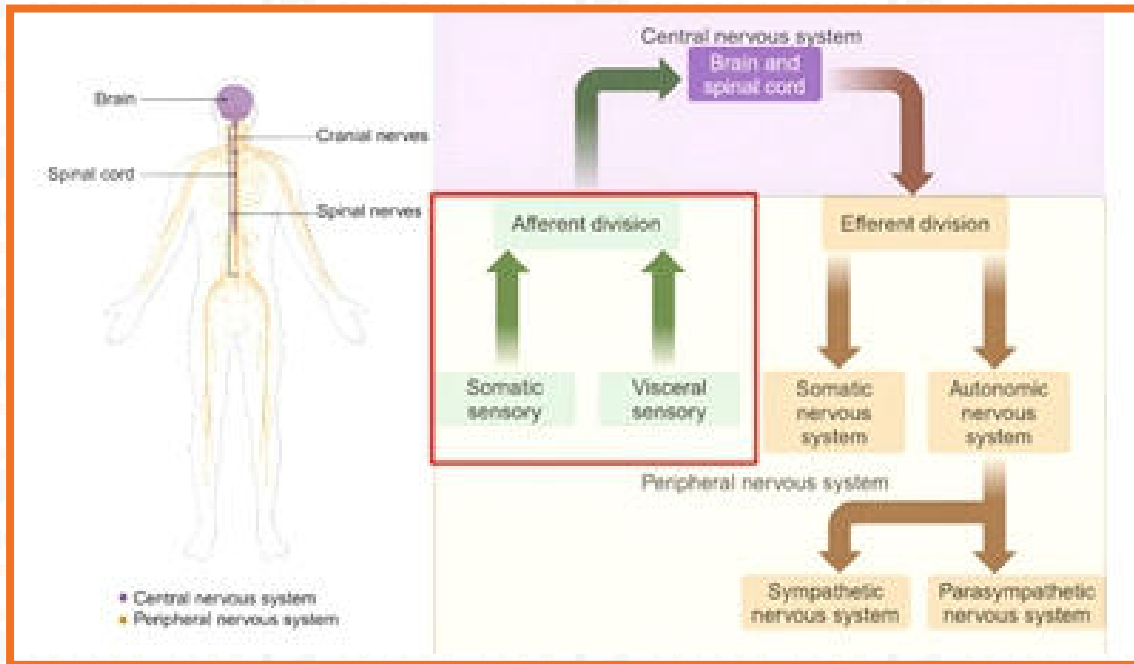
- **Receptors** are **sensory cells** that detect the energy of a stimulus.
- They are usually **specialized neurons** or **epithelial cells** found alone or grouped inside **sensory organs** (e.g., eyes, ears).





What is a Transducer?

- A transducer is a structure that converts signals from one form of energy into another.
- All receptors act as biological transducers by converting various forms of external/internal stimuli into electrical nerve impulses.



Five Categories of Receptors (Based on Energy Type Detected)

1. Mechanoreceptors

- Stimulated by physical deformation (mechanical energy).
- Respond to: pressure, touch, stretch, motion, sound.

2. Pain Receptors (Nociceptors)

- Detect stimuli that cause or might cause tissue damage.
- Perceived as pain.

3. Thermoreceptors

- Respond to heat or cold.
- Help regulate body temperature by signaling both surface and core temperatures.

4. Chemoreceptors

- Detect chemical stimuli:
- Osmoreceptors (in hypothalamus) monitor solute concentration in blood and trigger thirst when osmolarity rises.
- Olfactory receptors (in nasal epithelium) detect smell.
- Gustatory receptors (on tongue) detect taste.

5. Photoreceptors

- Detect light stimuli.
- Found in the eyes:
- Rods (low light, black-and-white vision).
- Cones (color vision, sharp detail)





Signal Flow:

1. Sensory Input:

- All signals from receptors are transmitted to the **central nervous system (CNS)**.

2. Processing:

- **Interneurons** in the CNS process and analyze the incoming information to decide on the proper response.

3. Effectors:

- **Effectors** (muscles or glands) receive commands from the CNS.
- They **produce a response** to the stimulus (e.g., muscle contraction, gland secretion).

The Path of a Message Transmitted to the CNS:

Central Nervous System (CNS):

- Made up of the **brain** and **spinal cord**.
- Receives incoming signals (sensory information) and produces **responses** by directing specific neurons to communicate with **effector organs** (muscles, glands).

Step-by-Step Pathway:

1. Detection by Sensory Receptors:

- **Specific sensors** detect a stimulus.
- These sensors can be:
- **Sensory neuron nerve endings** in organs.
- Or **specialized cells** that send signals to sensory neurons.

2. Transmission by Sensory Neurons:

- The **sensory neuron** generates a **nerve impulse**.
- This impulse travels **toward the CNS** (brain or spinal cord).
- Sensors in various body parts (**eyes, ears, muscles, skin, etc.**) use sensory neurons to deliver these messages.

3. Processing in the CNS:

- The **brain or spinal cord** receives the incoming message.
- The CNS **processes** the information and decides on an appropriate response.

4. Signal Sent to Effector via Motor Neurons:

- A **motor neuron** carries the response message from the CNS **to the effector organ**.
- **Effector organs** (muscles or glands) then carry out the response.

Overall Flow Summary:

Step	Key Players
Detect stimulus	Sensory receptors, sensory neuron nerve endings
Transmit to CNS	Sensory neurons
Process and decide response	CNS (brain, spinal cord)
Carry out response	Motor neurons → Effector organs (muscles, glands)





Key Terms Recap:

- **Sensory neuron** → carries signals to CNS.
- **Motor neuron** → carries commands from CNS to effectors.
- **Effector organ** → carries out the action (muscle contraction, gland secretion).

Neurons:

Definition:

- A neuron is the functional and structural unit of the nervous system.
- It is specialized to receive, process, and transmit electrical signals.

Main Parts of a Neuron:

1. Cell Body (Soma):

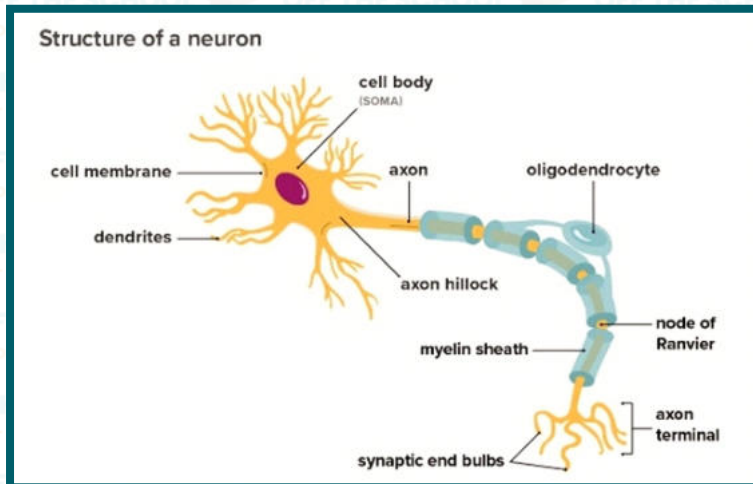
- Contains the **nucleus** and most organelles.
- Controls the **metabolic activities** of the neuron.

2. Dendrites:

- **Meaning:** (Greek dendron = tree).
- Branched projections extending from the cell body.
- Specialized to **receive signals** from other neurons or the external environment.
- Their **branched shape** provides a large surface area for receiving inputs.

3. Axon:

- **Meaning:** (Greek axon = axis).
- A **long, slender fiber** extending from the cell body, making neurons the **longest cells** in the body.
- The **cytoplasm** inside the axon is called **axoplasm**, and the membrane is called **axolemma**.
- **Function:** Conducts electrical signals **away** from the cell body to:
 - Other neurons, or
 - Effector organs (muscles, glands).



Supporting Cells: Neuroglia (Glial Cells):

Definition:

- Cells that support neurons structurally and functionally.
- Roles include:
 - Supplying nutrients.
 - Removing wastes.
 - Guiding axon migration.
 - Providing immune protection.





Types of Neuroglia:

Type	Location	Function
Schwann cells	Peripheral Nervous System (PNS)	Produce myelin in PNS.
Oligodendrocytes	Central Nervous System (CNS)	Produce myelin in CNS.

Neurons:

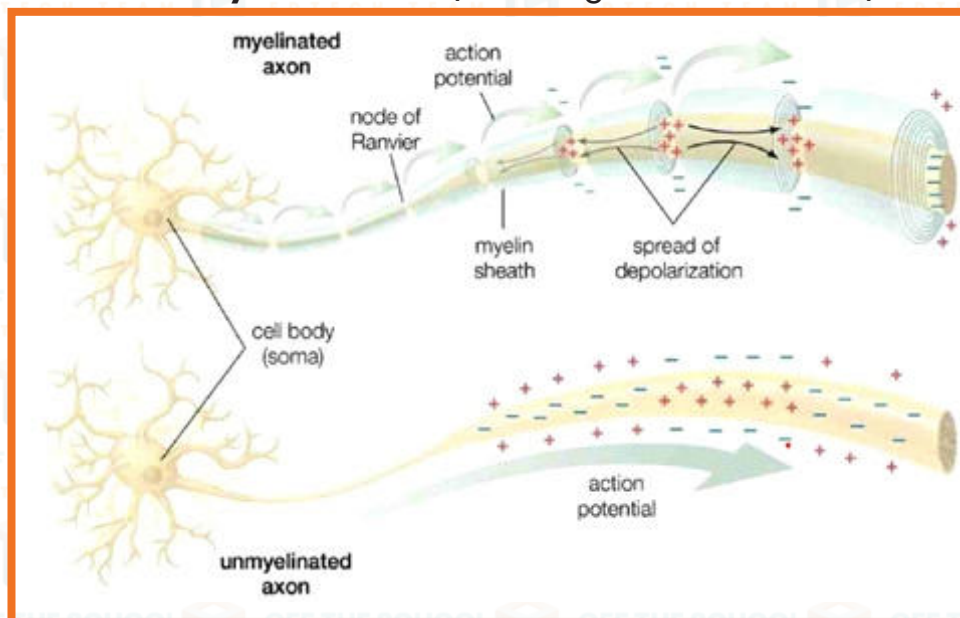
Definition:

Myelin and Axon Types:

- **Myelinated fibers** → Axons covered with **myelin sheath**.
- **Unmyelinated fibers** → Axons **without** myelin sheath.

Node of Ranvier (Neurofibril Nodes):

- The **gaps** between adjacent Schwann cells on a myelinated axon.
- Important for **saltatory conduction** (faster signal transmission).



Summary Table:

Neuron Part	Function
Cell body (soma)	Metabolic center, contains nucleus.
Dendrites	Receive incoming signals.
Axon	Transmit electrical signals away from soma.
Schwann cells	Produce myelin in PNS.
Oligodendrocytes	Produce myelin in CNS.
Node of Ranvier	Gaps enabling faster impulse conduction.





Types of Neurons (Functional Classification):

1. Sensory Neurons (Afferent Neurons):

- Found mainly in **sensory organs** (eyes, nose, skin, tongue, ears).
- Detect **chemical and physical inputs** like sound, heat, light.

Function:

Carry **sensory impulses** from **receptors** to the **central nervous system (CNS)**.

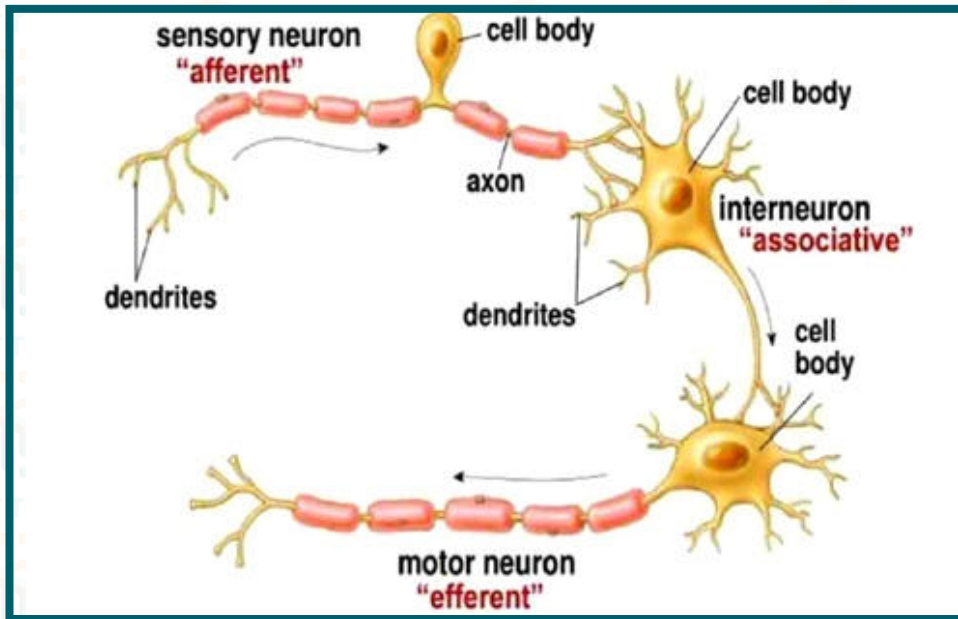
2. Motor Neurons (Efferent Neurons):

Function:

- Transmit **motor impulses** from the **CNS to effector organs** (muscles, glands).

Important for:

- Both voluntary movements (e.g., walking) and **involuntary** actions (e.g., heart beating).



3. Interneurons (Association Neurons):

- Act as **mediators** between sensory neurons, motor neurons, and the CNS.
- Function:
- Help ensure **smooth transmission** of signals within the CNS.
- Found mainly in the **brain and spinal cord**.

Reflex Action:

Definition:

- A **simple, automatic** response to a stimulus, called a **reflex** or **reflex action**.
- Involves a direct pathway from **sensory neuron** → **motor neuron** → **effector**.

Reflex Arc:

- The **pathway** along which impulses are transmitted from a receptor to an effector during a reflex.

Key components:

- **Sensory receptor** → detects stimulus.
- **Sensory neuron** → carries signal to CNS.
- **Motor neuron** → carries signal to effector.
- **Effector** → produces response (e.g., muscle contraction).

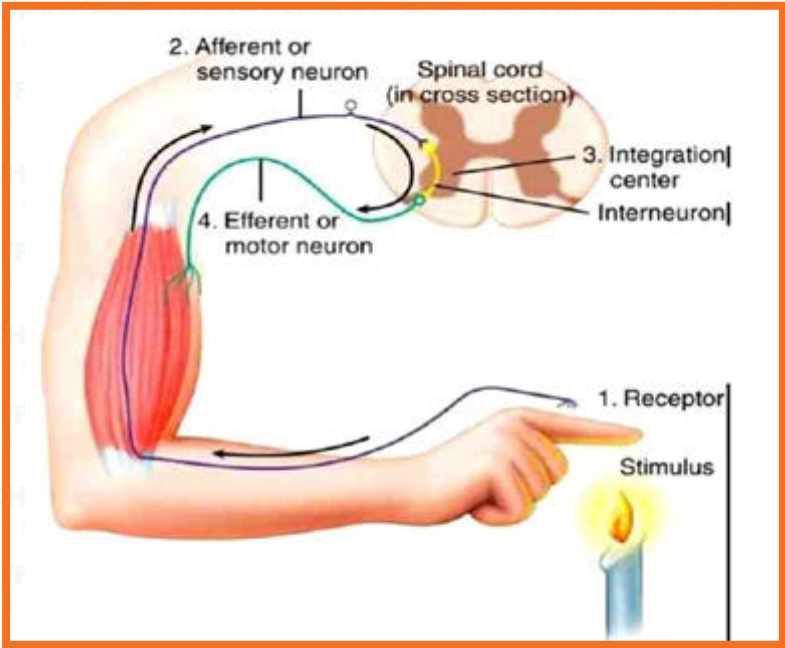


Examples of Human Reflexes:

- **Knee-jerk reflex** (patellar reflex).
- **Pain withdrawal reflex** (pulling hand away from a sharp or hot object).

Note

- Some reflexes (like the pain withdrawal reflex) involve **only sensory and motor neurons — interneurons (brain)** are not required for the reflex itself.
- However, other neural pathways do inform the brain afterward (e.g., feeling pain after reflex action)



Summary Table:

Neuron Type	Function
Sensory Neuron	Carries signals from receptors to CNS.
Motor Neuron	Carries signals from CNS to effectors (muscles, glands).
Interneuron	Connects sensory and motor neurons inside CNS for coordination.

Reflex Arc Components	Role
Receptor → Sensory Neuron → CNS → Motor Neuron → Effector	Automatic, quick response pathway (reflex).

Nerve Impulse:

Definition:

- A **nerve impulse** is an **electrical signal** that depends on the **flow of ions** across a neuron's membrane.
- The signal starts when there is a change in the **electrical gradient** (voltage) across the plasma membrane.



Generation and Transmission of Nerve Impulse:

Membrane Potential:

- All living cells maintain an **electrical charge difference** across their membrane (neurolemma).
- This difference creates an **electrical voltage gradient** known as the **membrane potential**.

Electrochemical Transmission:

- Nerve impulses are transmitted **electrochemically** through the neuron (dendrites or axon).
- **Voltage-gated protein channels** in the neurolemma open and close in response to changes in electrical voltage, allowing the impulse to travel.

Polarization of Neurolemma:

- The **inside** of the neurolemma is **negatively charged** relative to the outside.
- This polarization is due to the specific **distribution of ions**.

Ion Distribution in a Resting Neuron:

Ion Type	Location (Major Concentration)
Potassium (K^+)	High inside the cell (~30 times higher than outside).
Sodium (Na^+)	High outside the cell (~10 times higher than inside).
Negative ions (Cl^- , PO_4^{2-} , SO_4^{2-} , proteins)	Inside the cell (large, non-diffusible negative ions).

Sodium-Potassium Pump:

- Maintains ion distribution by actively transporting:
- Sodium (Na^+) out of the cell.
- Potassium (K^+) into the cell.
- Helps keep the resting membrane polarized.

Ion Channels:

- Membrane channels are ion-specific ☒ they allow only certain ions to pass while restricting others.

Summary Table:

Key Concept	Details
Membrane potential	Electrical charge difference across neurolemma.
Polarization	Inside negative, outside positive (due to ion distribution).
Sodium-potassium pump	Maintains Na^+ outside, K^+ inside using active transport.
Voltage-gated channels	Open/close in response to voltage changes, allowing impulse transmission.





Resting Membrane Potential (RMP):

Definition:

- Resting membrane potential is the electrical potential difference across the neuron's membrane when the neuron is not stimulated (at rest).
- It has a value of -70 millivolts (mV).
- The negative sign indicates that the inside of the neuron is more negative compared to the outside.

Key Points about RMP:

Membrane permeability:

- Membrane is slightly permeable to Na^+ and K^+ .
- More permeable to K^+ (potassium) than Na^+ (sodium).

Ion distribution:

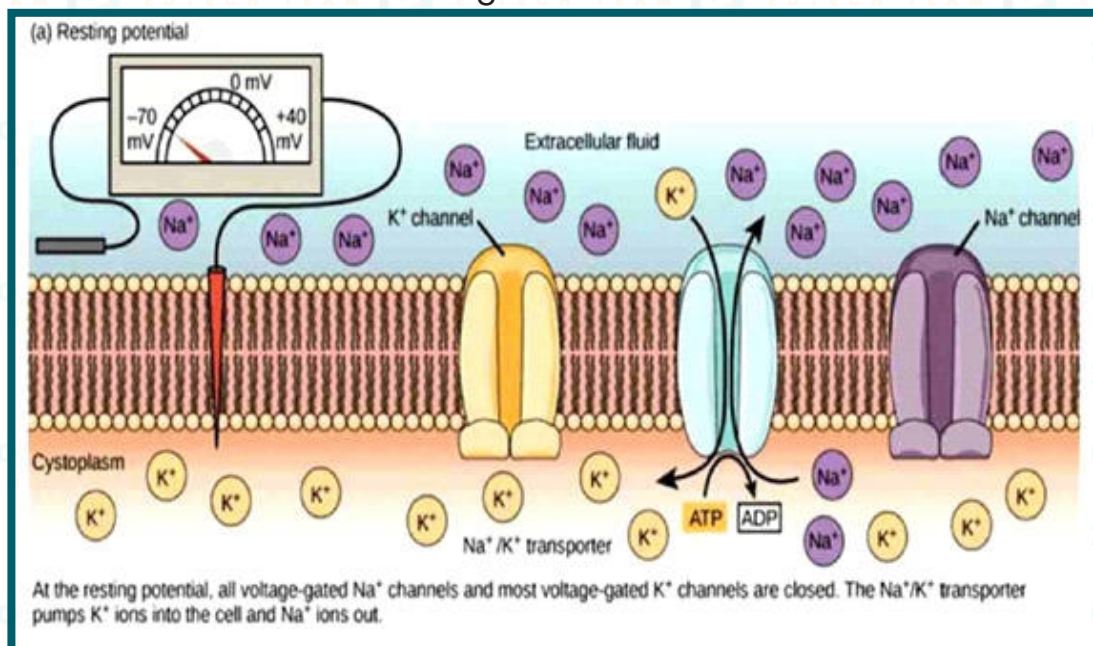
- K^+ (potassium) $\rightarrow$ high inside the cell.
- Na^+ (sodium) $\rightarrow$ high outside the cell.
- Cl^- (chloride) $\rightarrow$ tends to stay outside, repelled by negative proteins inside.

Potassium leakage channels:

- Allow K^+ to diffuse out faster than Na^+ leaks in.
- This net loss of positive ions leaves the inside more negative.

Sodium-potassium pump:

- Maintains RMP by actively transporting:
- 3 Na^+ out.
- 2 K^+ in.
- This creates and maintains the negative interior.



Changes in Membrane Potential:

Depolarization:

- When the membrane potential becomes less negative than RMP.
- Moves toward zero or even becomes positive.

Hyperpolarization:

- When the membrane potential becomes more negative than RMP.

Neurons are excitable $\rightarrow$ they respond to stimuli by altering membrane potential.



Summary of RMP:

Condition	Charge
Inside cell (at rest)	Negative (due to K^+ movement & proteins)
Outside cell	Positive (due to Na^+ & Cl^- accumulation)
RMP value	~ -70 mV

Action Membrane Potential (AMP):

Definition:

- Action potential is a brief, rapid change in membrane potential triggered by a stimulus, causing the inside of the membrane to become positive.

Steps in Action Potential

1. Threshold stimulus:

- A stimulus strong enough to trigger action potential (if too weak $\rightarrow$ sub-threshold stimulus, no response).

2. Depolarization:

- Na^+ channels open $\rightarrow Na^+$ rushes in.
- Inside becomes positive (can reach up to $+50$ mV).

3. Repolarization:

- Na^+ channels close.
- K^+ channels open $\rightarrow K^+$ moves out.
- Inside becomes negative again.

4. Return to Resting State:

- Restores the -70 mV resting potential.
- Takes about 10–30 milliseconds.

Summary Table: Action Potential Process

Phase	What Happens
Threshold stimulus	Stimulus strong enough to trigger impulse.
Depolarization	Na^+ influx $\rightarrow$ inside becomes positive.
Repolarization	K^+ efflux $\rightarrow$ inside returns to negative.
Recovery	Restores resting potential ($\sim 10\text{--}30$ ms).

Depolarization:

Definition:

- Depolarization refers to the process where the membrane potential becomes less negative (or even positive) due to the influx of sodium ions (Na^+) into the neuron.





Key Features:

- Occurs when a graded potential reaches the threshold level (~ -55 mV) from the resting membrane potential (~ -70 mV).
- This threshold stimulus must be strong enough to trigger an action potential.

Mechanism

1. Stimulus Trigger:

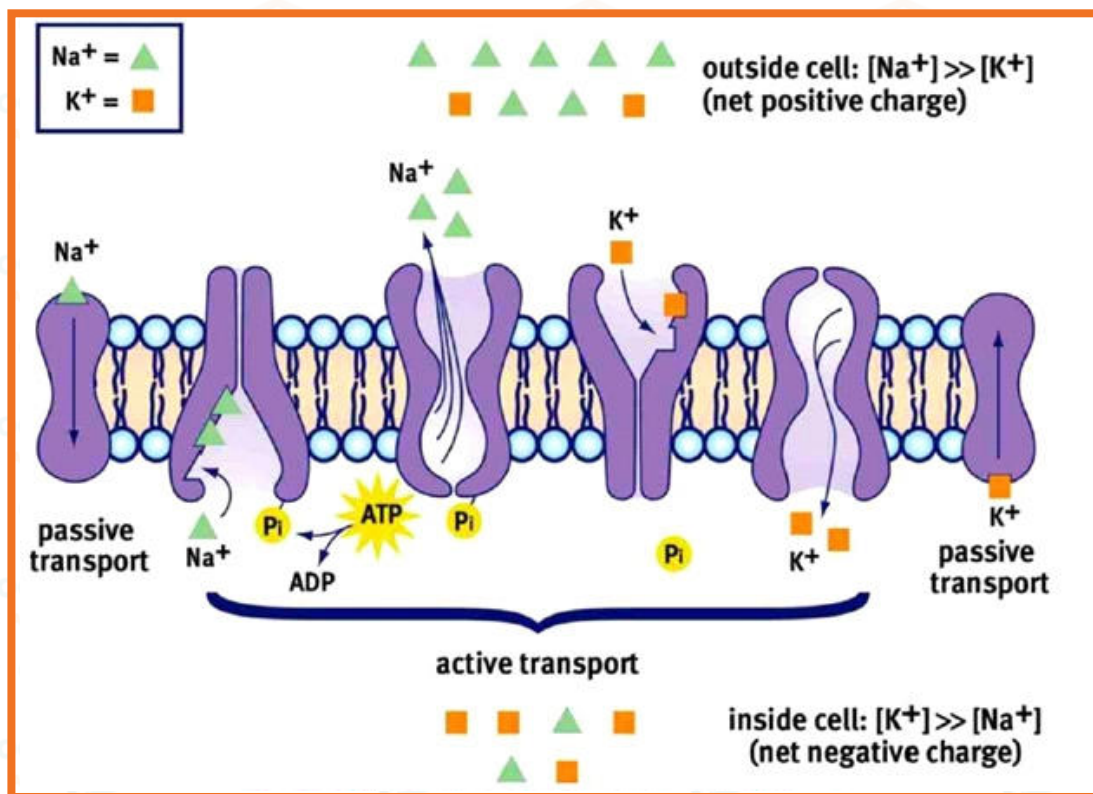
- A stimulus causes the voltage-gated sodium (Na^+) channels in the neurolemma to open.

2. Sodium Influx:

- Na^+ ions rush into the cell, about 10 times more than during the resting state.
- This influx reduces the negativity inside the cell, causing the membrane potential to shift toward positive.

3. Change in Membrane Potential:

- Rapid shift from negative (resting) $\rightarrow$ positive (action potential).
- Once the inside becomes positive, the Na^+ channels close or become inactivated.



All-or-None Principle

- Action potential follows an all-or-none rule:
- If the stimulus is below threshold $\rightarrow$ no action potential generated.
- If the stimulus is at or above threshold $\rightarrow$ full action potential generated across the entire neurolemma.





Summary Table:

Phase	What Happens
Resting potential	Na ⁺ channels closed, inside negative (~ -70 mV).
Threshold stimulus	~ -55 mV, triggers opening of Na ⁺ voltage-gated channels.
Depolarization	Na ⁺ influx $\rightarrow$ inside becomes positive (up to $+50$ mV).
End of depolarization	Na ⁺ channels close/inactivate, stopping Na ⁺ influx.

Repolarization:

Definition:

- Repolarization is the phase during which the neuron restores its resting membrane potential after depolarization.
- The inside of the neuron returns to a negative charge relative to the outside.

Key Features

- Occurs after depolarization, when the inside of the axon has become positively charged.
- Involves outward movement (efflux) of K⁺ (potassium) ions.

Mechanism

1. Sodium Channels Close:

- After depolarization, voltage-gated sodium (Na⁺) channels close, stopping the Na⁺ influx.

2. Potassium Channels Open:

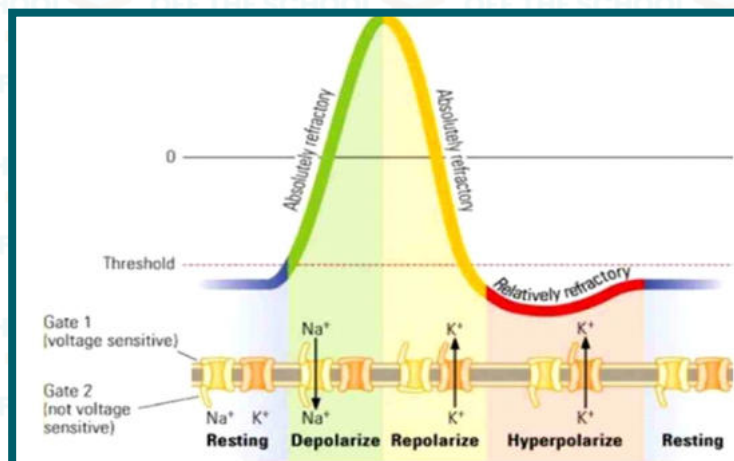
- Voltage-gated potassium (K⁺) channels open.
- K⁺ ions flow out of the axoplasm to the extracellular space.

3. Restoring Negative Potential:

- Efflux of K⁺ leads to a loss of positive charge inside, restoring the negative membrane potential (~ -70 mV).

4. Re-establishing Ionic Balance:

- Repolarization resets the conditions, allowing the neuron to be ready for the next action potential.





Summary Table:

Phase	What Happens
End of depolarization	Na^+ channels close, stopping sodium entry.
Repolarization	K^+ channels open $\rightarrow$ K^+ efflux $\rightarrow$ inside becomes negative again.
Restored state	Membrane potential returns to resting level (~ -70 mV).

Importance:

- Repolarization ensures the neuron:
- Restores its electrical balance.
- Prepares for the next nerve impulse.
- Maintains proper neural signaling and prevents continuous firing.

Refractory Period:

Definition:

- The refractory period is the brief time after a nerve impulse during which the neuron cannot immediately respond to another stimulus.
- It allows the membrane to reset and prepare for the next nerve impulse.

Key Features:

- Lasts approximately 2 milliseconds.
- Ensures one-way transmission of nerve impulses along the neuron.
- Prevents continuous or overlapping signals.

Mechanism:

1. Restoration of Ionic Balance:

- After the action potential, the sodium-potassium ATPase pump actively restores:
- 3 Na^+ ions moved out.
- 2 K^+ ions moved in.

2. Against Concentration Gradient:

- This ion movement is against the concentration gradients of both Na^+ and K^+ , requiring ATP energy.

3. Resetting Resting Membrane Potential:

- The pump re-establishes the original ion distribution:
- High Na^+ outside.
- High K^+ inside.
- This resets the resting membrane potential (~ -70 mV).

Importance of Refractory Period:

- Prevents immediate re-stimulation of the neuron.
- Ensures that each action potential is separate and distinct.
- Allows unidirectional propagation of the nerve impulse.
- Prepares the neuron for the next signal.





Summary Table:

Aspect	Details
Duration	~2 milliseconds
Ion transport	Na ⁺ pumped out, K ⁺ pumped in (using ATP)
Purpose	Restore resting state; prepare for next impulse
Role of sodium-potassium pump	Maintains ionic gradients across the membrane

Hyperpolarization:

Definition

● **Hyperpolarization** is a state where the **membrane potential becomes more negative** than the resting potential (~ -70 mV).

Key Features of Hyperpolarization:

1. Cause:

- Occurs after **repolarization**, when **potassium (K⁺) ions continue to move out** or **chloride (Cl⁻) ions** enter the cell through their respective channels.
- This causes the membrane potential to dip below the resting value, e.g., **from -70 mV to -75 mV**.

2. Restoration:

- The membrane potential returns to normal when:
- **All voltage-gated ion channels close.**
- The **sodium-potassium pump** re-establishes the resting potential.

Velocities of Nerve Impulse:

Definition:

- The **velocity of nerve impulse conduction** refers to the speed at which action potentials travel along neurons.

Factors Affecting Nerve Impulse Velocity

1. Axon Diameter:

- The **larger the axon diameter**, the **faster the nerve impulse**.
- Larger diameter reduces **electrical resistance**.
- **Electrical resistance** is inversely proportional to **cross-sectional area** (larger diameter = lower resistance = faster conduction).

2. Myelination:

- **Myelinated axons** conduct impulses **faster** than **unmyelinated axons**.
- Action potentials are produced **only at the Nodes of Ranvier** (gaps between Schwann cells).
- This allows **saltatory conduction**, where the impulse "jumps" from node to node, skipping the myelinated sections, speeding up transmission.



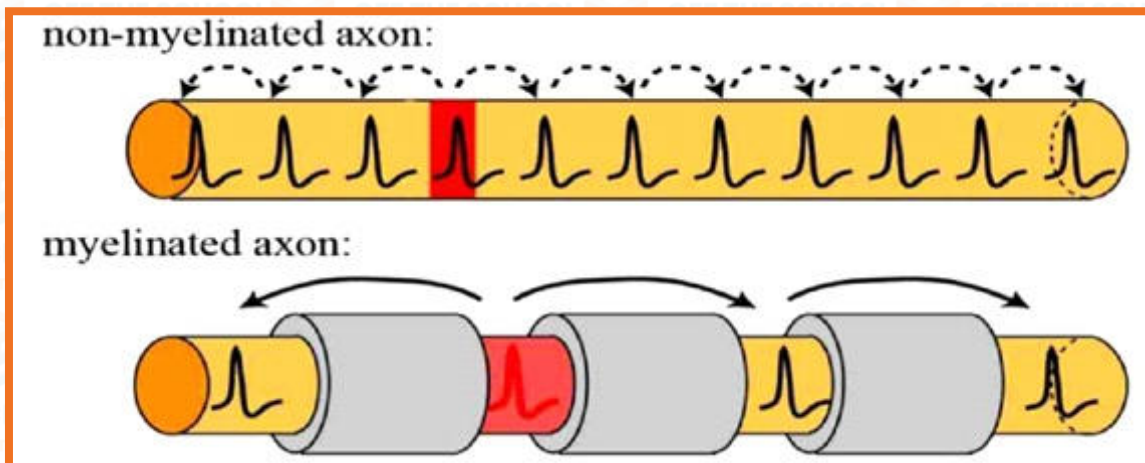


Myelinated vs Non-Myelinated Axons:

Axon Type	Speed of Conduction	Mechanism
Myelinated	Up to 150 m/s	Saltatory conduction: action potential jumps between nodes.
Non-myelinated	Slower (several cm/s)	Action potential propagates along entire axon.

Summary of Nerve Impulse Speed:

- **Axon Diameter:** Larger = Faster transmission.
- **Myelination:** Myelinated = Faster (due to saltatory conduction).
- **Speed Range:** From **several cm/s** in thin axons to **150 m/s** in giant, myelinated axons.



Synapse:

Definition:

- A **synapse** is the junction that enables **communication between a neuron and another cell** (another neuron, muscle cell, gland, or sensory receptor).
- It controls the transmission of nerve impulses across gaps between cells.

Structure of Synapse:

Pre-synaptic Cell:

- The neuron whose **axon transmits the action potential** to the synapse.

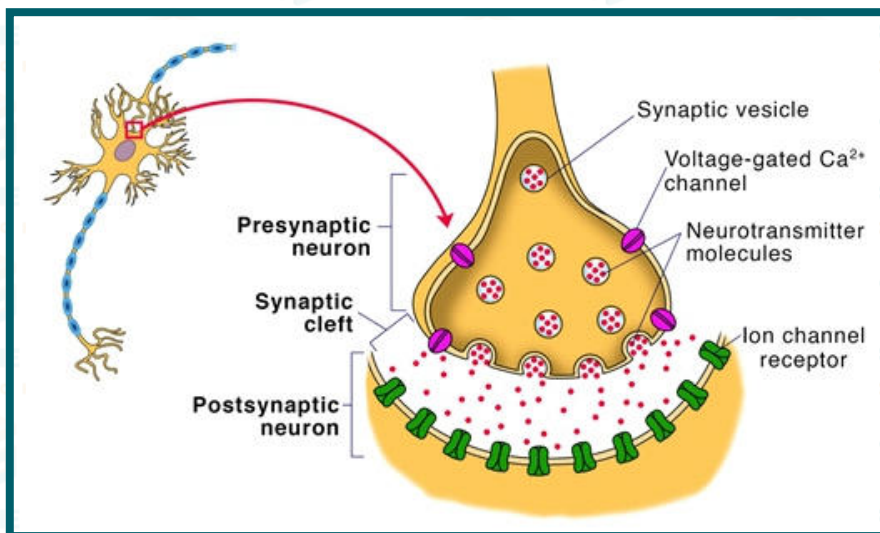
Post-synaptic Cell:

- The cell that **receives the signal** from the pre-synaptic neuron.

Synaptic Cleft:

- A small gap between the pre-synaptic and post-synaptic cells through which the neurotransmitters travel.





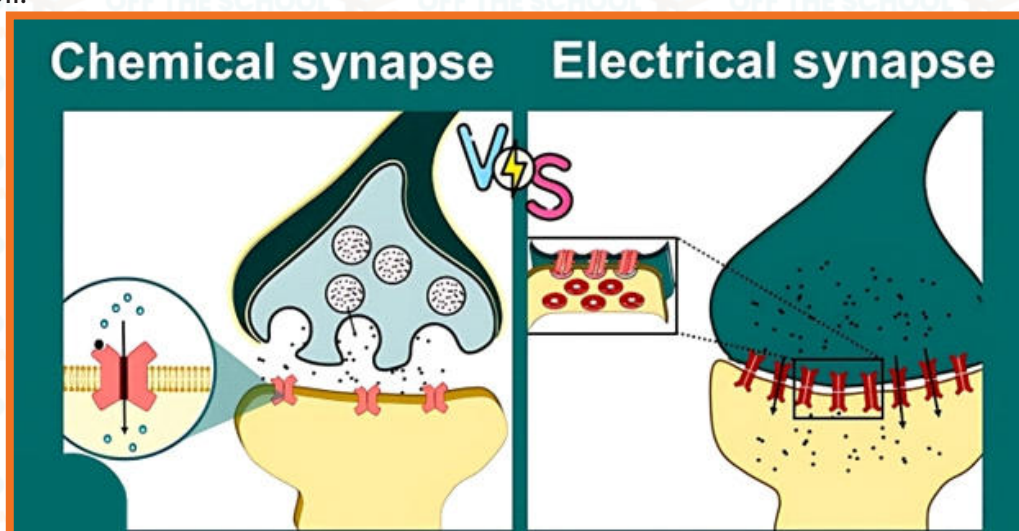
Types of Synapses:

1. Electrical Synapse:

- **Direct Cytoplasmic Connection:** Formed by **gap junctions** between the pre- and post-synaptic neurons.
- **Transmission:** Impulses are transmitted **without delay** and **without loss of signal strength**.
- **Characteristics:**
 - Rapid communication, important for movements like reflexes.
 - **Common in invertebrates**, less in vertebrates.

2. Chemical Synapse:

- **Common in vertebrates.**
- The **presynaptic axon terminal** contains **synaptic vesicles** packed with neurotransmitters.
- **Steps of Transmission:**
 - **Action Potential** reaches the pre-synaptic axon terminal.
 - This triggers the opening of **voltage-gated calcium (Ca^{2+}) channels**.
 - **Ca^{2+} influx** causes synaptic vesicles to fuse with the plasma membrane.
 - **Neurotransmitters** are released by **exocytosis** into the synaptic cleft.
 - Neurotransmitters bind to **receptors** on the post-synaptic membrane.
 - This binding can either **excite** (depolarize) or **inhibit** (hyperpolarize) the post-synaptic cell.





Mechanism of Synaptic Transmission:

1. Neurotransmitter Release:

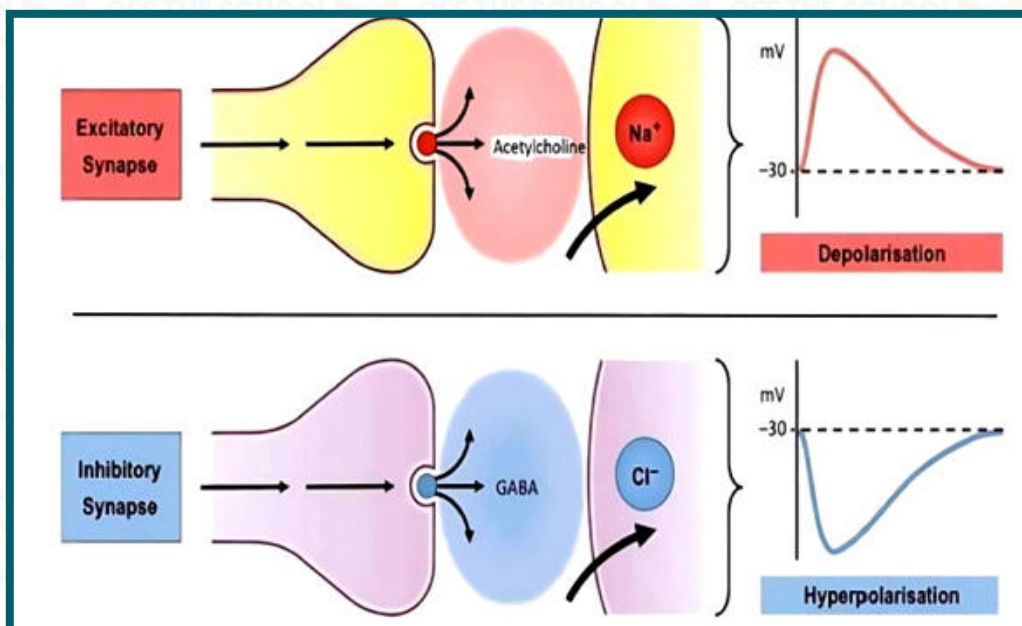
- Action potential causes the influx of calcium ions (Ca^{2+}).
- This triggers the fusion of synaptic vesicles with the membrane, releasing neurotransmitters into the synaptic cleft.

2. Post-synaptic Effect:

- The binding of neurotransmitters to receptors can either:
- Excite the post-synaptic cell (by depolarizing the membrane).
- Inhibit the post-synaptic cell (by hyperpolarizing the membrane).

3. Termination of Signal:

- Neurotransmitters must be removed from the synaptic cleft to stop the signal:
- Enzymatic digestion (breakdown of neurotransmitters).
- Reuptake into the presynaptic neuron.
- Uptake by glial cells.



Feature	Electrical Synapse	Chemical Synapse
Communication Type	Direct cytoplasmic connections (gap junctions)	Release of neurotransmitters
Speed	Fast (no delay)	Slower (requires neurotransmitter release)
Signal Strength	No loss of signal strength	Signal may be modified (excite/inhibit)
Common Location	Invertebrates and some CNS regions	Vertebrates, CNS and PNS
Direction	Bi-directional	Unidirectional



Classification of Neurotransmitters

Neurotransmitters can be broadly classified as **excitatory** or **inhibitory**, depending on their effect on the postsynaptic membrane.

i. Excitatory Neurotransmitters:

1. Mechanism of Action:

- At an **excitatory synapse**, neurotransmitter receptors control ion channels that allow **Na⁺** to enter the cell and **K⁺** to leave the cell.
- This causes **depolarization**, which is a change in membrane potential that brings the membrane closer to the threshold for an action potential.
- The electrical change caused by neurotransmitter binding is called **Excitatory Postsynaptic Potential (EPSP)**.

2. Examples of Excitatory Neurotransmitters:

- Acetylcholine (ACh):**
 - One of the most common neurotransmitters.
 - Can be **excitatory or inhibitory** depending on the type of receptor.
- Biogenic Amines** (derived from amino acids):
 - Epinephrine** and **Norepinephrine**: Also function as hormones; involved in the **autonomic nervous system**.
 - Dopamine**: Involved in movement, attention, and learning; **Parkinson's disease** is associated with low dopamine levels.
 - Serotonin**: Regulates **sleep, mood, attention, and learning**; an imbalance is linked to **depression** and other mood disorders.

ii. Inhibitory Neurotransmitters:

1. Mechanism of Action:

- At an **inhibitory synapse**, neurotransmitter binding causes **hyperpolarization** of the postsynaptic membrane.
- This occurs by opening ion channels that increase membrane permeability to:
 - K⁺** (potassium) which rushes **out** of the cell.
 - Cl⁻** (chloride) which enters the cell.
- This makes the membrane more **negative** than the resting potential, making it **more difficult** for an action potential to occur.
- The electrical change caused by neurotransmitter binding is called **Inhibitory Postsynaptic Potential (IPSP)**.

2. Examples of Inhibitory Neurotransmitters:

- Gamma-Aminobutyric Acid (GABA)**: The most common inhibitory neurotransmitter in the brain.
- Glycine**: Inhibitory neurotransmitter mainly in the **spinal cord**.
- Glutamate and Aspartate**: Although generally excitatory, they can have inhibitory effects under certain conditions.

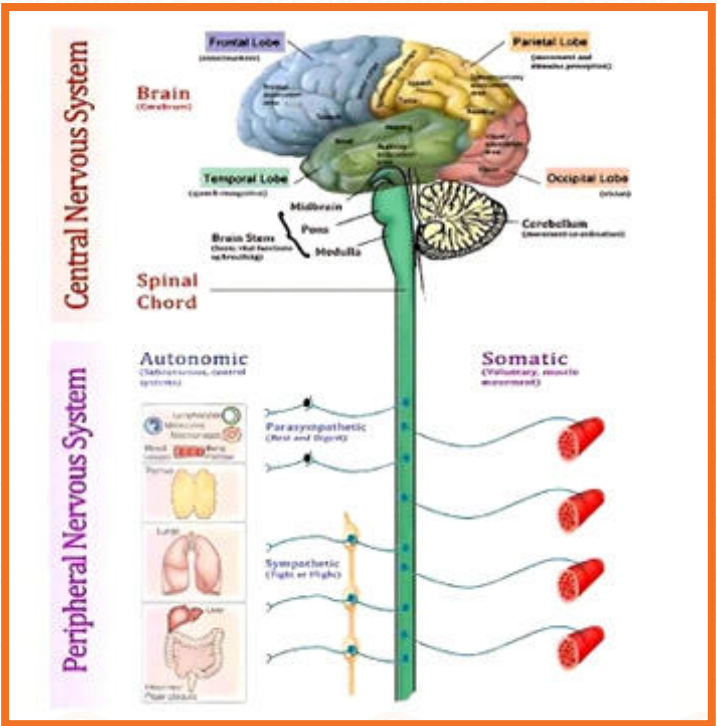


Summary:

Type of Neurotransmitter	Effect	Examples
Excitatory	Causes depolarization and action potential	Acetylcholine, Epinephrine, Norepinephrine, Dopamine, Serotonin
Inhibitory	Causes hyperpolarization , making action potential less likely	GABA, Glycine, Glutamate, Aspartate

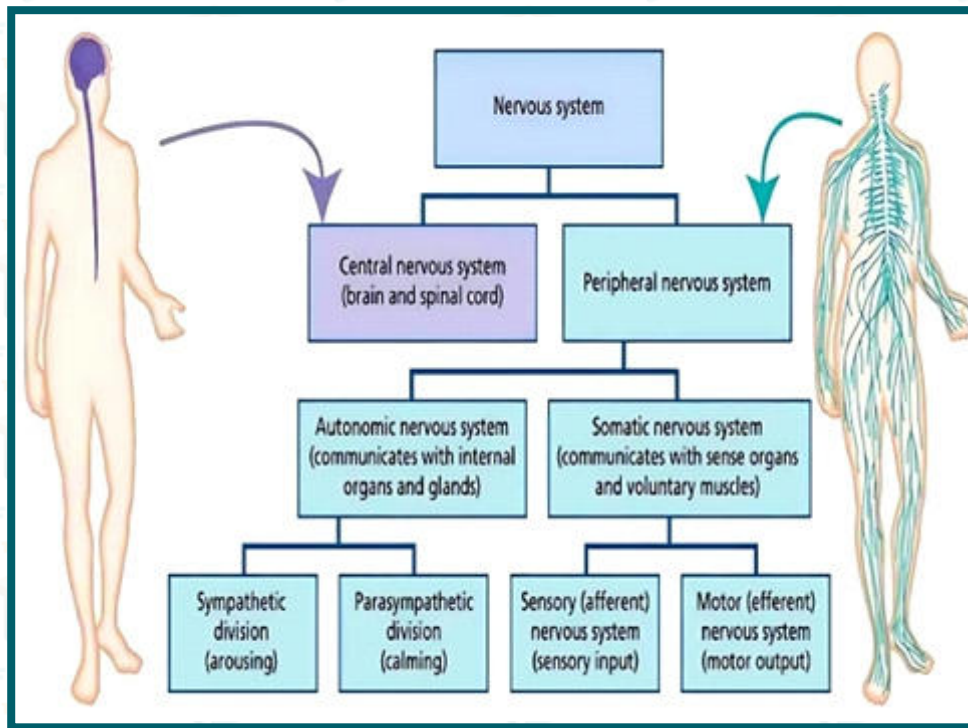
Basic Organization of the Human Nervous System:

The human nervous system is divided into two main parts:



1. Central Nervous System (CNS):

- **Components:**
- **Brain**
- **Spinal Cord**
- **Location:**
- The CNS lies in the **central** region of the body, specifically along the **midline**.
- The **brain** is located within the skull, and the **spinal cord** runs along the vertebral column.
- **Function:**
- The CNS is responsible for processing and integrating sensory information.
- It coordinates responses to stimuli, regulates functions, and is involved in higher cognitive functions like thinking, memory, and decision-making.



2. Peripheral Nervous System (PNS):

- **Components:**

- Nerves that extend from the CNS to the rest of the body.

- **Location:**

- The PNS lies in the **peripheral** regions of the body, extending outward from the CNS to limbs and organs.

- **Function:**

- The PNS serves as a communication system between the CNS and the rest of the body.
- It transmits sensory input to the CNS and carries motor output from the CNS to the muscles and glands.

Architecture of Brain and Spinal Cord (Central Nervous System – CNS):

The **CNS** is the integrating center of the nervous system, where sensory information is processed, and thoughts and decisions are made. It is composed of the **brain** and **spinal cord**, which are both protected in multiple ways.

Protection of CNS:

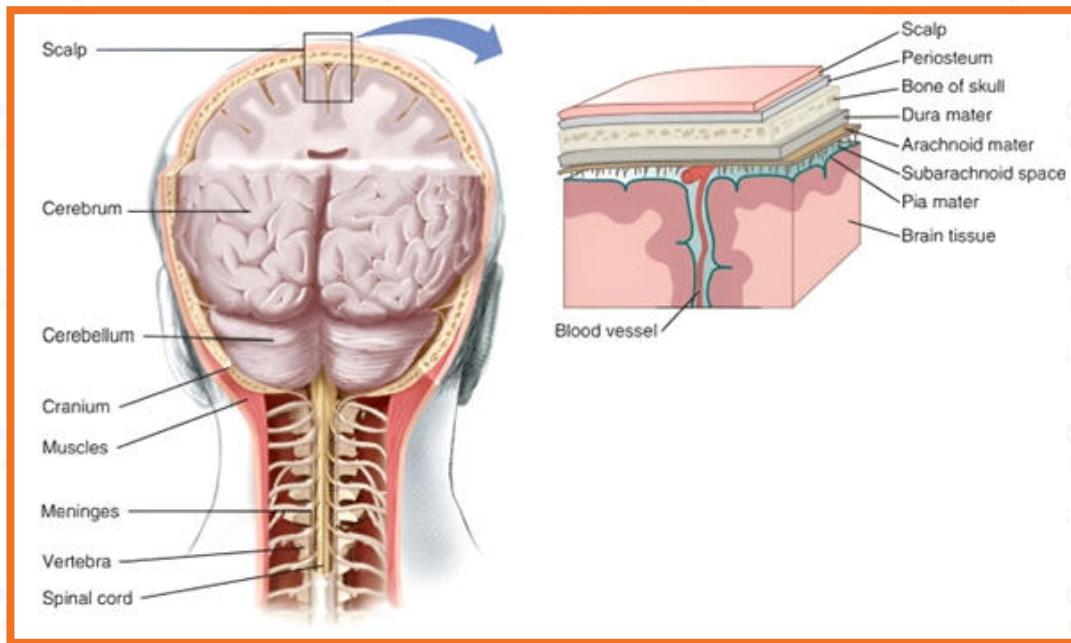
1. Bony Armor:

- **Skull:** Protects the brain.
- **Vertebral Column:** Protects the spinal cord.

2. Meninges:

- A triple layer of connective tissue that provides additional protection to the CNS.
- **Dura Mater:** The outermost layer, next to the cranium.
- **Arachnoid Mater:** The middle layer.
- **Pia Mater:** The innermost layer, next to the nervous tissue.
- **Cerebrospinal Fluid (CSF):** Found between the arachnoid and pia mater, it cushions and protects the brain and spinal cord.





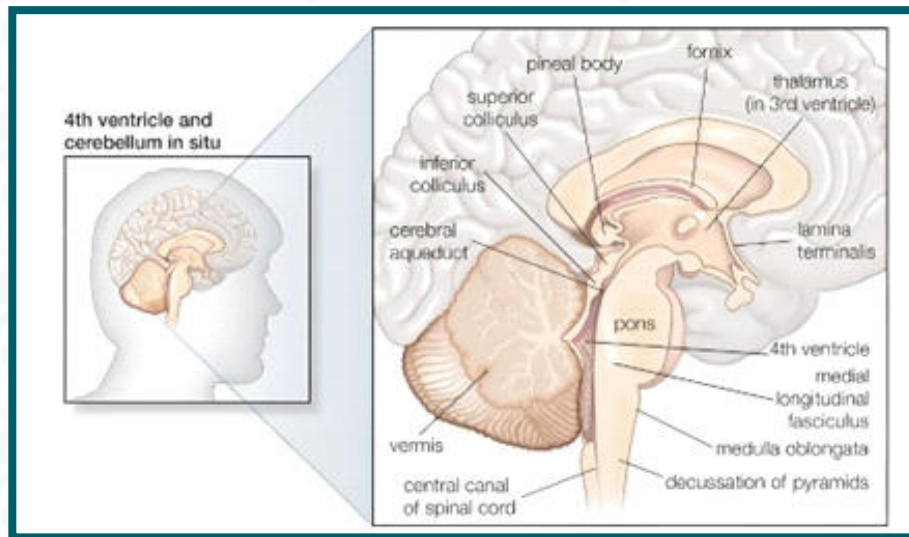
Brain Structure:

The brain is the primary component of the CNS and is housed within the skull. It is divided into three main regions: Forebrain, Midbrain, and Hindbrain.

1. Forebrain:

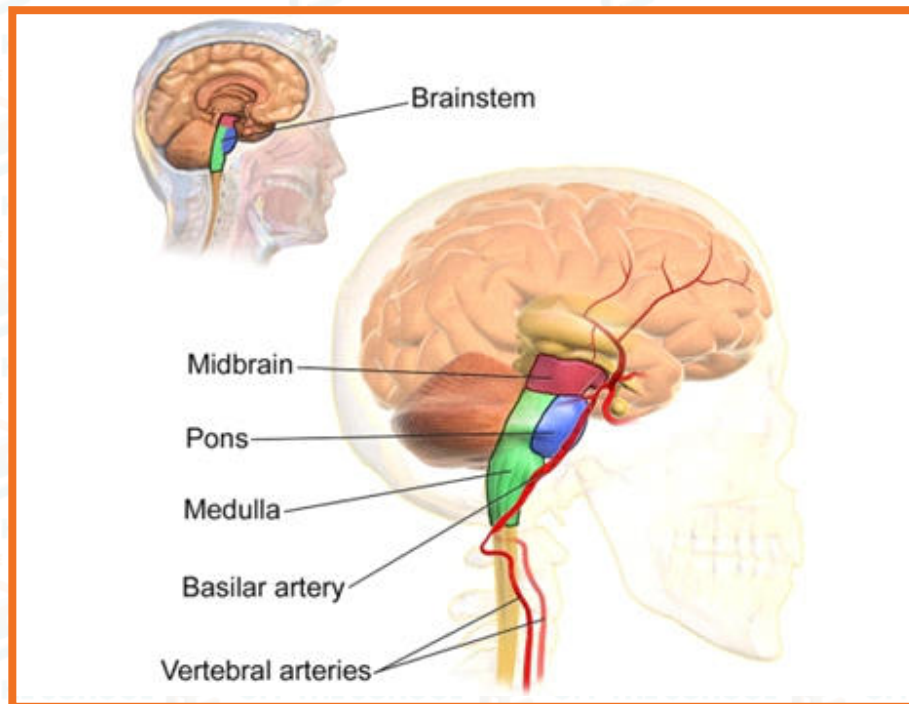
- **Telencephalon:**
 - Divided into two **cerebral hemispheres** (cerebrum).
 - The hemispheres are connected by the **corpus callosum** (a band of axons).
- **Cerebral Cortex:**
 - Outer grey matter responsible for higher cognitive functions.
 - Divided into four lobes:
 - **Frontal Lobe:** Decision making, reasoning, voluntary movements.
 - **Parietal Lobe:** Sensory processing.
 - **Occipital Lobe:** Visual processing.
 - **Temporal Lobe:** Auditory processing and memory.
- **It has three areas:**
 - **Sensory Area:** Receives input from body parts.
 - **Association Area:** Responsible for higher mental activities (e.g., reasoning, memory).
 - **Motor Area:** Controls body movements.
- **Diencephalon:**
 - Includes the Thalamus and Limbic System.
- **Thalamus:** Major integrating center for sensory and motor information.
- **Limbic System:**
 - **Hypothalamus:** Regulates bodily functions like hunger, thirst, body temperature, and the autonomic nervous system.
 - **Amygdala:** Controls emotions like fear, rage, and pleasure.
 - **Hippocampus:** Involved in the formation of long-term memory and learning.





2. Midbrain:

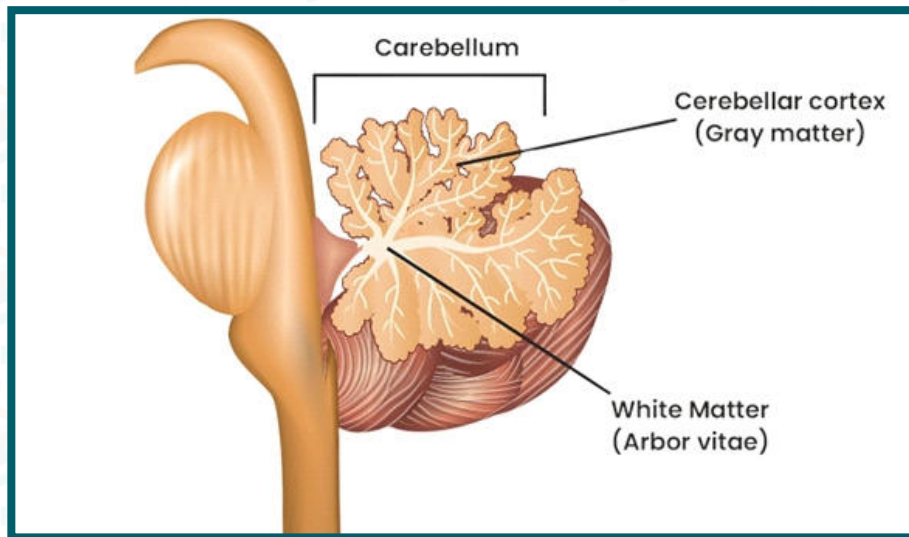
- The **Midbrain** is a reduced structure in humans but plays an important role in relaying signals.
- Contains the **reticular formation**, which extends through the medulla, pons, and midbrain, involved in sleep, arousal, and reflexes.



3. Hindbrain:

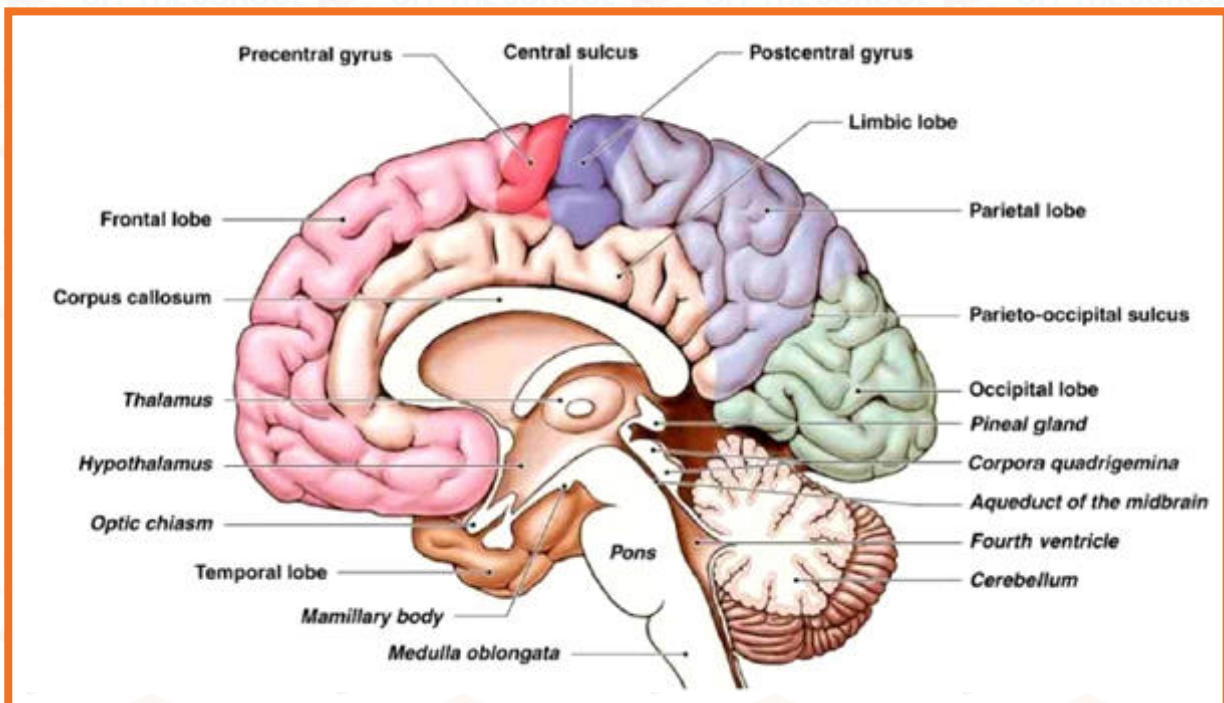
- The **Hindbrain** consists of the **Medulla**, **Pons**, and **Cerebellum**.
- **Medulla**: Controls autonomic functions like breathing, heart rate, and blood pressure.
- **Pons**: Influences transitions between sleep and wakefulness and regulates breathing patterns.
- **Cerebellum**: Coordinates movement and balance, processing sensory information from joints, muscles, and other systems.





Brainstem:

- The **Brainstem** connects the spinal cord to the brain and controls vital involuntary functions.
- It consists of three parts:
 - **Medulla Oblongata:** Controls heart rate, breathing, and other autonomic functions.
 - **Pons:** Connects the medulla to higher brain centers, influencing breathing and sleep.
 - **Midbrain:** A relay center for sensory and motor pathways, playing a role in visual and auditory processing.





Spinal Cord:

The spinal cord is a long neural cable extending from the brain down through the vertebral column (backbone). It is protected by both the vertebral column and the meninges, which are layers of connective tissue also covering the brain.

Structure of the Spinal Cord:

1. Inner Zone (Grey Matter):

- Primarily composed of **cell bodies of interneurons, motor neurons, and neuroglia** (supporting cells).
- Grey matter** is the site of synaptic connections and processing of information.

2. Outer Zone (White Matter):

- Contains the **axons of sensory neurons** (in the **dorsal column**) and **motor neurons** (in the ventral column).
- White matter** is responsible for transmitting signals up and down the spinal cord. It may also contain the **dendrites** of some nerve cells.

3. Function of the Spinal Cord:

Information Highway: It relays messages between the brain and the rest of the body.

Reflexes: The spinal cord plays a key role in reflexes, which are rapid, involuntary movements in response to a stimulus. Reflexes occur when **sensory neurons** transmit information to **motor neurons** in the spinal cord, resulting in a quick motor response.

Cranial and Spinal Nerves:

Peripheral Nervous System (PNS): Composed of **cranial nerves, spinal nerves**, and their associated **ganglia**.

Cranial Nerves: There are **12 pairs** of cranial nerves, originating in the brain, and innervating the head and upper body. Most cranial nerves are **mixed (sensory and motor)**, except for the **olfactory** and **optic nerves**, which are sensory only.

Spinal Nerves: There are **31 pairs** of spinal nerves, originating from the spinal cord and innervating the rest of the body.

Organization of Spinal Nerves:

1. Sensory and Motor Components:

Sensory Neurons: The axons of sensory neurons enter the **dorsal surface** of the spinal cord, forming the **dorsal root**.

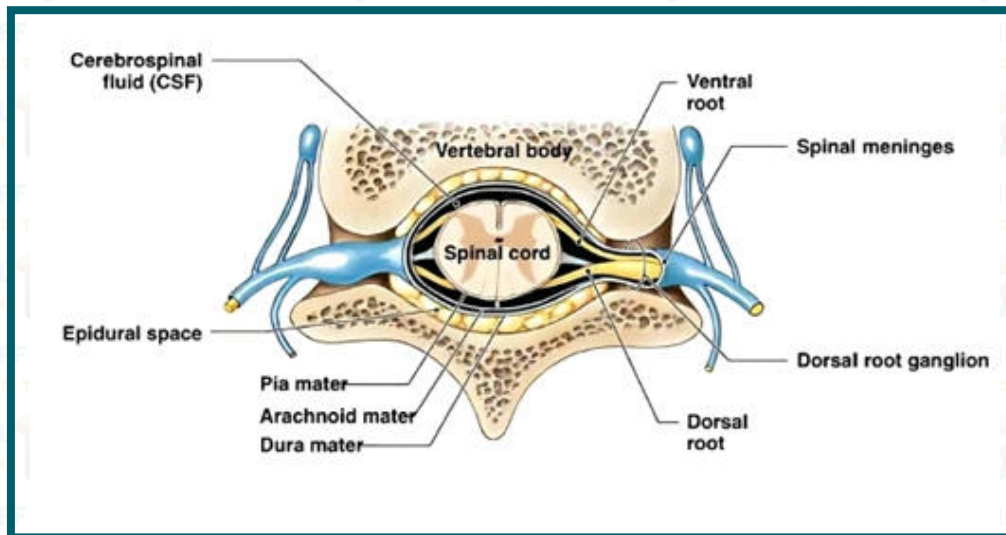
Motor Neurons: The axons of motor neurons exit from the **ventral surface** of the spinal cord, forming the **ventral root**.

2. Ganglia:

The **cell bodies** of sensory neurons are grouped together in the **dorsal root ganglia**, located outside the spinal cord.

The **cell bodies** of motor neurons are located **inside the spinal cord**, and are not in ganglia.



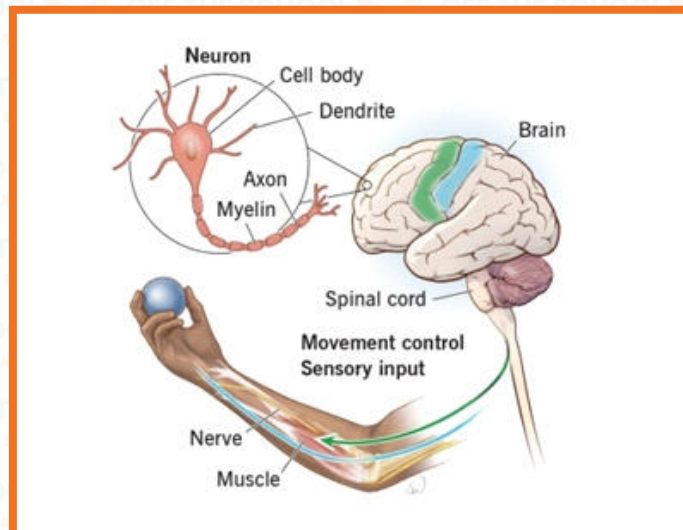


Somatic and Autonomic Nervous System:

The **motor portion** of the **peripheral nervous system** is divided into two major components: the **somatic nervous system** and the **autonomic nervous system**.

Somatic Nervous System:

- **Voluntary Control:** Often considered voluntary because it is subject to **conscious control**.
- **Function:**
 - Stimulates the **skeletal muscles** to contract.
 - Involved in **conscious movements** and **reflexes** that do not require conscious thought.



Autonomic Nervous System:

- **Involuntary Control:** Controls involuntary functions in the body, including the regulation of internal organs.
- **Functions:**
 - Regulates **smooth muscle, cardiac muscle, and glands**.
 - Involves control of the **gastrointestinal, cardiovascular, excretory, and endocrine systems**.
 - Generally operates **involuntarily**.

The autonomic nervous system has two subdivisions that are anatomically, physiologically, and chemically distinct:





1. Sympathetic Nervous System:

● **Function:** Prepares the body for **stressful or energetic activities** (e.g., fighting, escaping, or giving a speech).

● Known as the **"Fight-or-Flight"** response system.

● **Nerve origin:** The sympathetic nervous system arises from the **thoracic** and **lumbar** regions of the spinal cord.

● **Effects:**

● Increases **heart rate**.

● Dilates air passages in the lungs.

● Dilates pupils.

● Mobilizes energy for physical action.

2. Parasympathetic Nervous System:

● **Function:** Dominates during **maintenance activities** that occur when the body is at rest.

● Often called the **"Rest and Digest"** system.

● **Nerve origin:** Arises from **cranial nerves** from the brain and **sacral nerves** from the spinal cord.

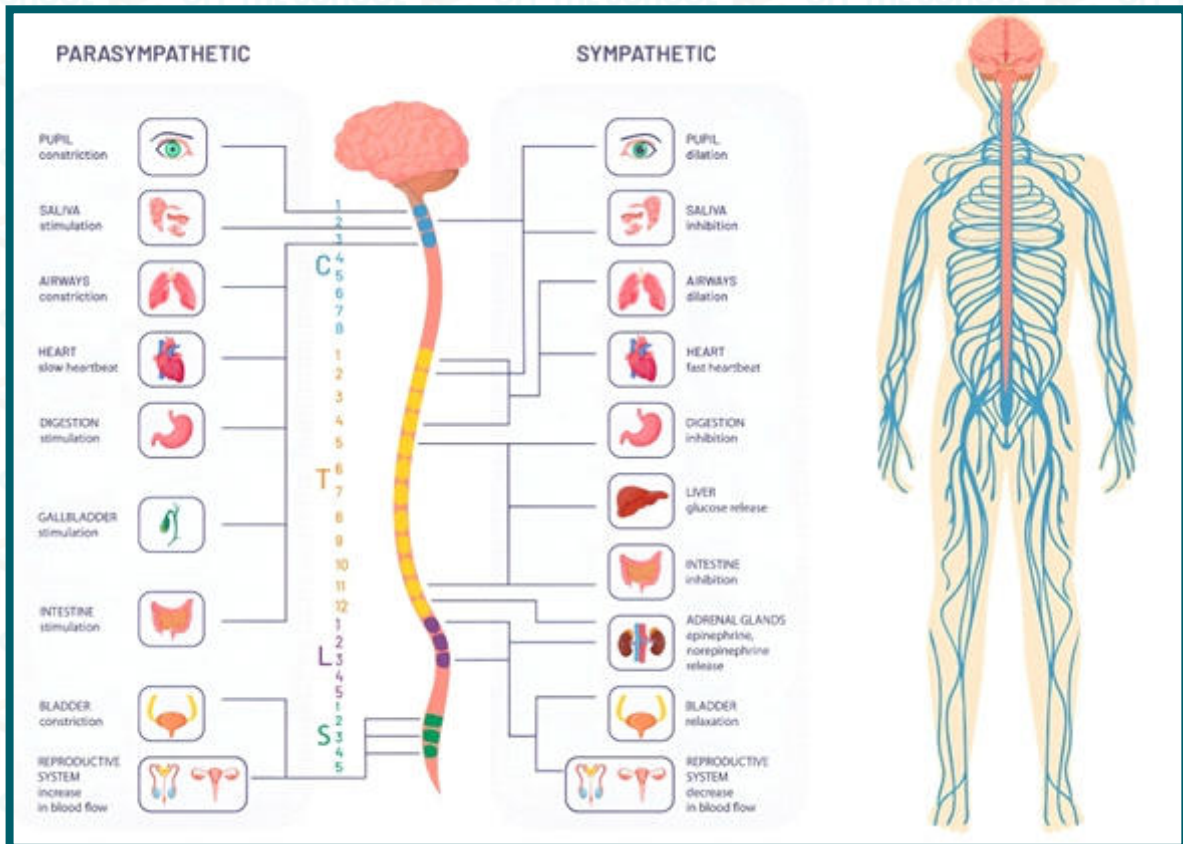
● **Effects:**

● Slows **heart rate**.

● Stimulates digestion and gastrointestinal activity.

● Constricts air passages in the lungs.

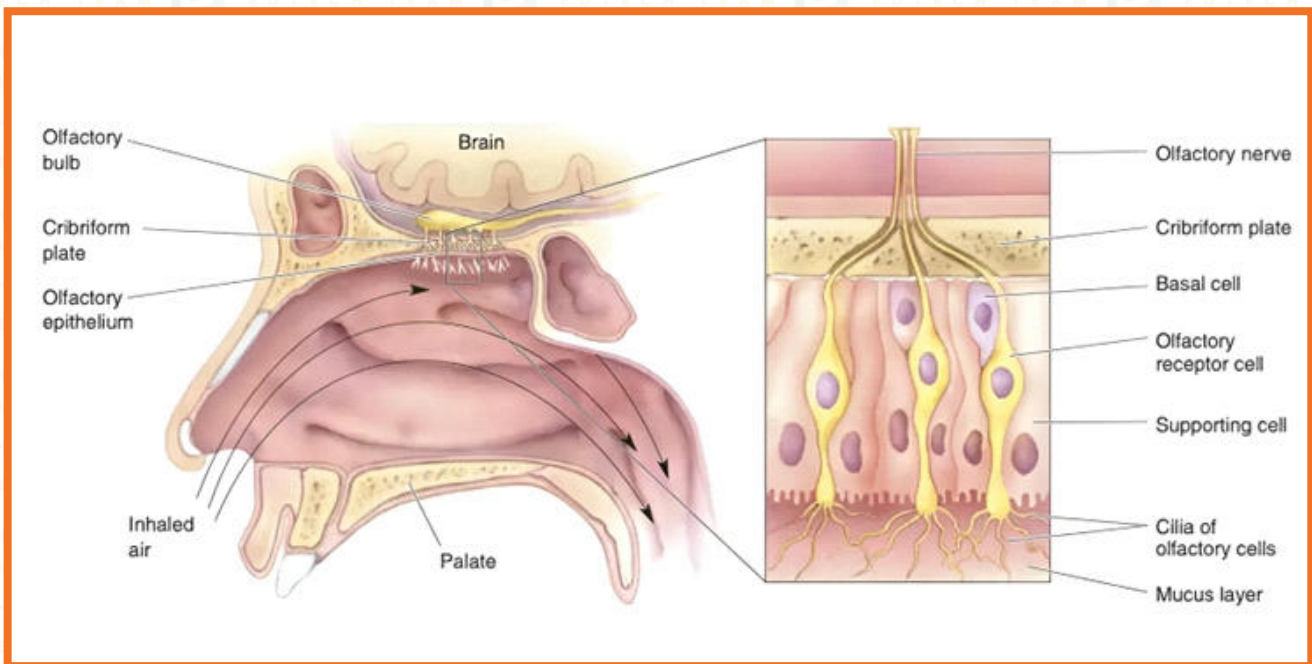
● Conserves energy and supports maintenance of body functions.





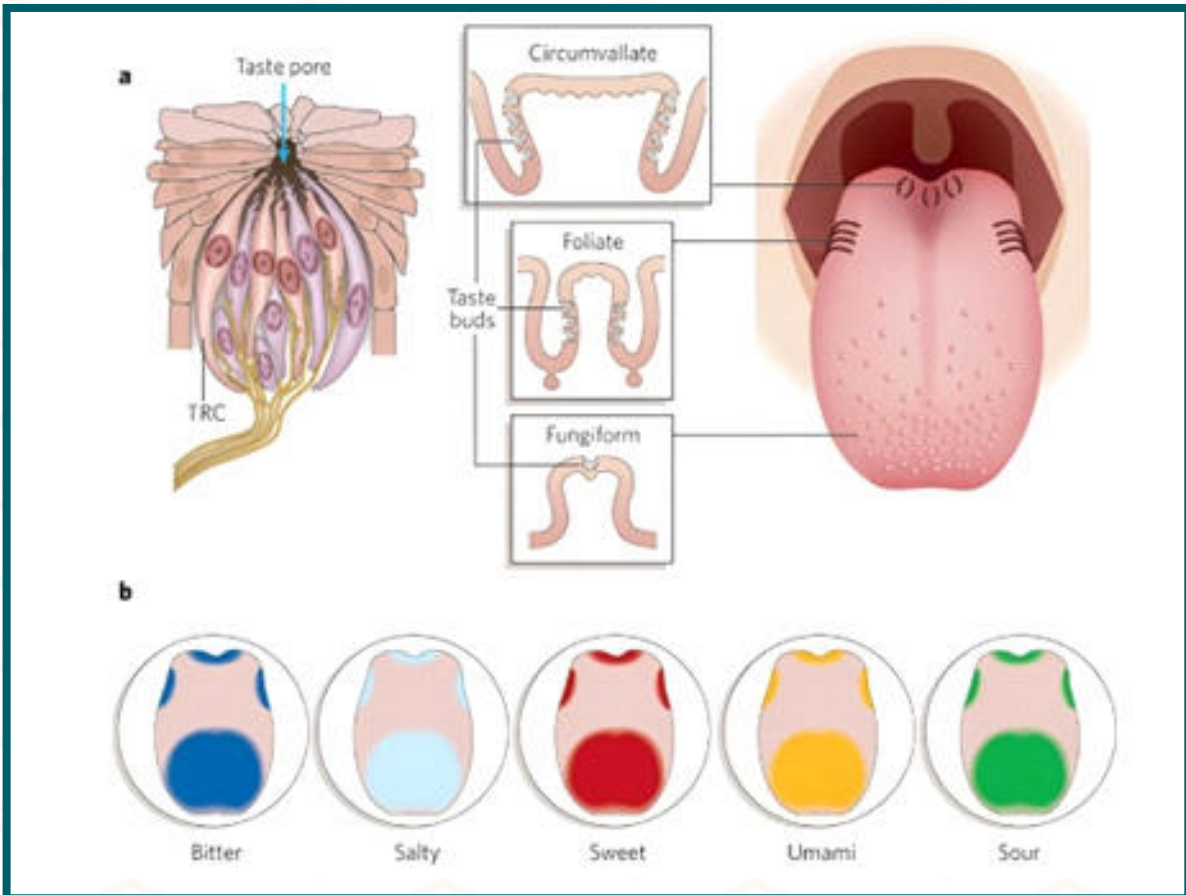
Smell Receptors (Olfactory Receptors):

- **Olfactory Neurons:** The dendrites of olfactory neurons end in **cilia** that project into the **nasal mucosa**. These neurons have axons that directly project into the cerebral cortex.
- **Receptor Molecules:** When **odorous substances** enter the nasal cavity, they **bind to specific receptor molecules** on the plasma membrane of the olfactory cilia.
- **Receptor Diversity:** There are approximately **1000 different types of receptor proteins** embedded in the olfactory cilia, each specialized to bind a specific molecule.
- **Signal Transmission:** Once a receptor protein binds its corresponding molecule, it stimulates the **olfactory neuron** to send a signal to the brain, resulting in the perception of smell.



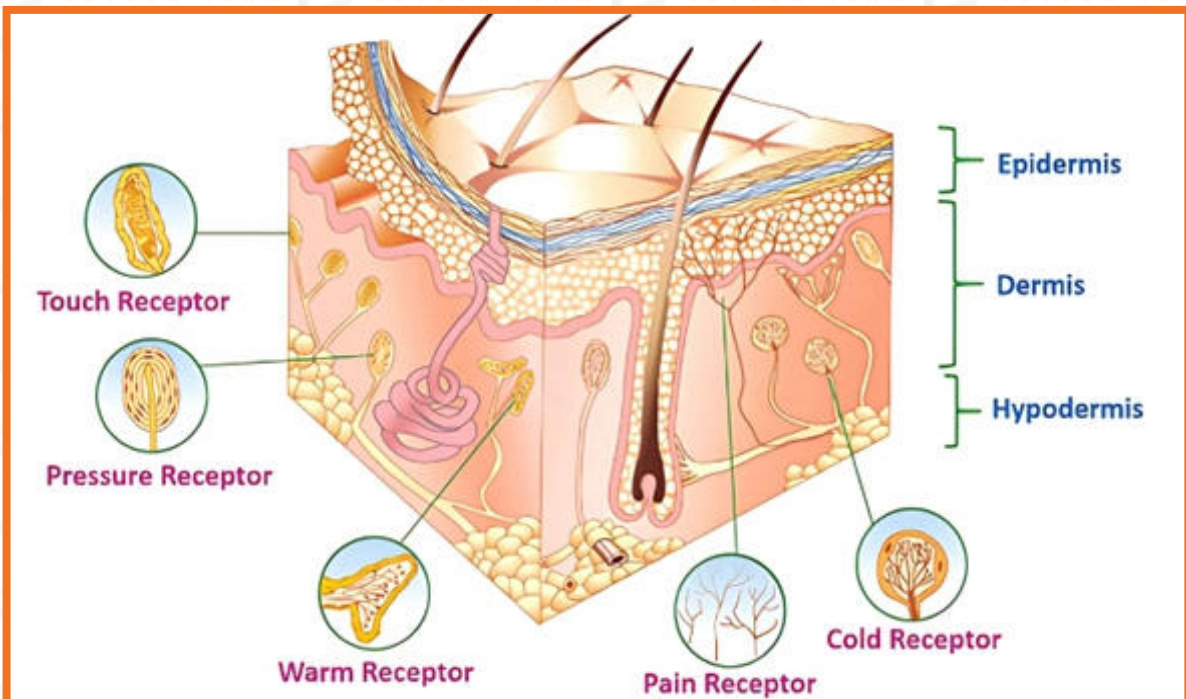
Taste Receptors (Gustation):

- **Taste Buds:** Taste receptor cells are modified epithelial cells organized into **taste buds**. Humans have around **10,000 taste buds**.
- **Location:** Taste buds are mainly found on the surface of the **tongue** and are associated with **papillae**, which are nipple-like projections on the tongue.
- **Taste Categories:** Humans recognize five basic tastes, though the structure of taste buds does not differentiate between them. The five recognized tastes are:
 1. **Sweet**
 2. **Sour**
 3. **Salty**
 4. **Bitter**
 5. **Umami (savory)**



Sensory Receptors in Human Skin:

Human skin contains various types of **mechanoreceptors** that are specialized to detect different forms of **physical contact**, contributing to the sense of **touch**. These receptors are found in the **dermis** and the **subcutaneous tissues**. The receptors are classified as **phasic** and **tonic** based on their response characteristics.





Phasic Receptors:

- **Function:** These receptors are responsive to **changes** in stimuli, such as **onset** or **offset** of touch.

1. Hair Follicle Receptors:

- **Location:** Around hair follicles.
- **Function:** Detects the movement of hair, providing sensations of light touch.

2. Meissner's Corpuscles:

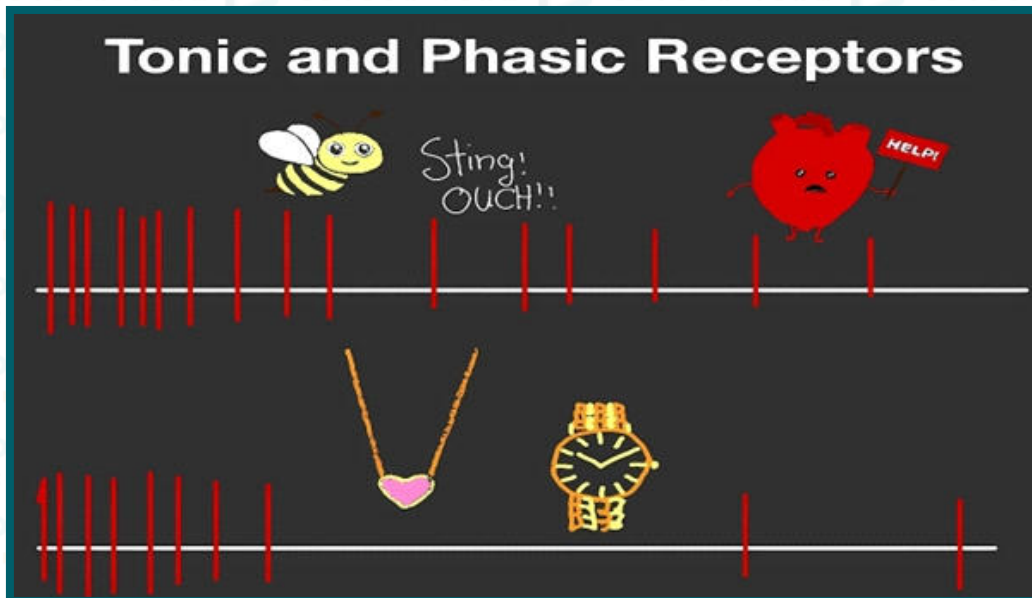
- **Location:** Found on **hairless skin surfaces** such as **fingers, palms, and nipples**.
- **Function:** Sensitive to **light touch** and **vibration**. These receptors respond to changes in pressure.

Tonic Receptors:

- **Function:** These receptors provide a continuous signal regarding the **duration** and **intensity** of stimuli.

1. Ruffini's Corpuscles:

- **Location:** In the **dermis**.
- **Function:** Detects **stretch** and **pressure**, contributing to the perception of skin deformation and sustained pressure.



2. Merkel's Disks (Touch Dome Endings):

- **Location:** Near the **surface of the skin**.
- **Function:** Responds to **steady pressure** and **texture**, helping to detect fine touch and pressure.

Deep Pressure Sensation:

1. Pacinian Corpuscles:

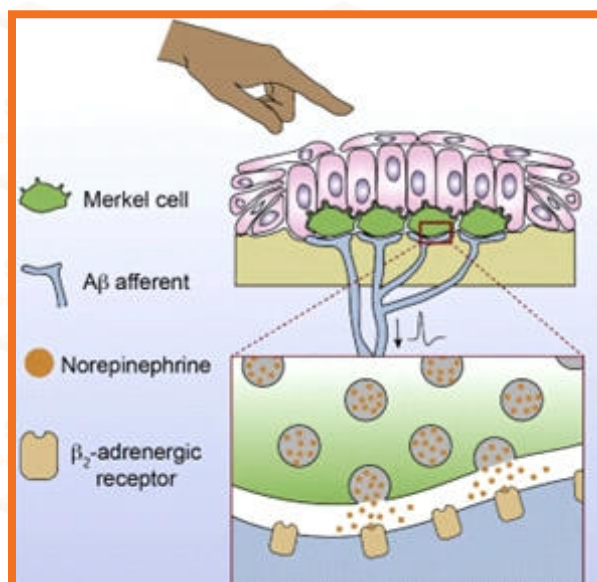
- **Location:** Deep in the skin and subcutaneous tissues.
- **Function:** Detects **deep pressure** and **vibration**. These receptors are sensitive to rapid changes in pressure.

Nociceptors

- **Location:** Found as **naked dendrites** in the **epidermis**.
- **Function:** Sensitive to **noxious stimuli**, including **tissue damage** and **pain**.

Nociceptors detect harmful stimuli that could cause injury, triggering a pain response.





Effect of Drugs on Nervous Coordination:

- **Pain Relief Drugs:** Drugs like **morphine** and **Demerol** block synapses in pain pathways of the **brain** and **spinal cord**, effectively reducing pain perception. The brain can also modulate pain using its natural narcotic-like substances called **endorphins**.
- **Endorphins:** These are naturally occurring substances in the brain that can block pain perception, especially in critical situations like **combat** or escaping from danger, allowing the body to function until the emergency passes.

Narcotic Drugs:

1. Heroin

- A **white powder** derived from the **morphine** alkaloid in **opium poppy**.
- **Effects:** Provides **euphoria**, anti-anxiety, and **pain-relieving properties**.
- **Addiction:** Highly addictive and commonly abused for its **euphoric** effects.

2. Cannabis (Marijuana)

- Derived from the **Cannabis sativa** plant.
- **Effects:** Causes mild **euphoria**, altered **vision**, and impaired **judgment**.
- **Forms:** Commonly consumed as **marijuana** (dried flowering tops) or **hashish** (resin).

3. Nicotine

- An alkaloid found in **tobacco**.
- **Mechanism:** Binds to specific receptors on postsynaptic neurons in the brain, which normally bind the neurotransmitter **acetylcholine**.
- **Effects:** Stimulates **heart rate**, increases **blood pressure**, and enhances **muscular activity**.

4. Alcohol

- A **depressant** that interferes with the brain's communication pathways.
- **Effects:** Alters mood, behavior, and cognitive functions, making it harder to think clearly and move with coordination.





5. Inhalants:

- **Volatile substances** found in household items such as **spray paints, markers, glues, and cleaning fluids.**

- **Effects:** Have **psychoactive** (mind-altering) properties when inhaled, which can be harmful to both the brain and body.

Drug Addiction and Drug Tolerance:

Drug Addiction:

- **Definition:** Drug addiction is a **neurobiological disorder** that impacts the **brain** and **behavior**, leading to the loss of the ability to resist the urge to use drugs.

- **Substances Involved:**

- Illegal drugs (e.g., **cocaine**).
- Legal substances such as **alcohol, nicotine, marijuana**, and certain

medications.

- **Process:**

- Addiction often starts with experimentation. People may initially take drugs because of how they feel (**euphoria** or relaxation).

- **Repeated misuse** of drugs alters brain chemistry and changes how the brain works.

- This leads to a **loss of self-control** and an increased **desire** to continue using the drugs.

Drug Tolerance:

- **Definition:** Drug tolerance occurs when repeated exposure to a drug results in a **decreased effect**, meaning a person needs to take **larger doses** of the substance to achieve the same effect.

- **Common Drugs Involved:**

- **Nicotine** and **caffeine** are examples of substances where tolerance develops over time.

- **Effect:** As tolerance builds, the body's **response to the drug weakens**, requiring higher doses to produce the same impact.

Disorders of the Nervous System and Diagnostic Techniques:

Classification of Nervous Disorders

Nervous system disorders can be categorized into several types, including:

- **Vascular** disorders
- **Infections** disorders
- **Structural** disorders
- **Functional** disorders
- **Degenerative** disorders

Vascular Disorders of the CNS:

Vascular disorders involve abnormalities in the **blood circulation** of the nervous system. These conditions typically affect the brain's blood supply and can lead to serious consequences.

Stroke:

- **Definition:** A stroke occurs when there is an interruption in the **blood supply to the brain**, leading to damage of brain tissue.





Risk Factors:

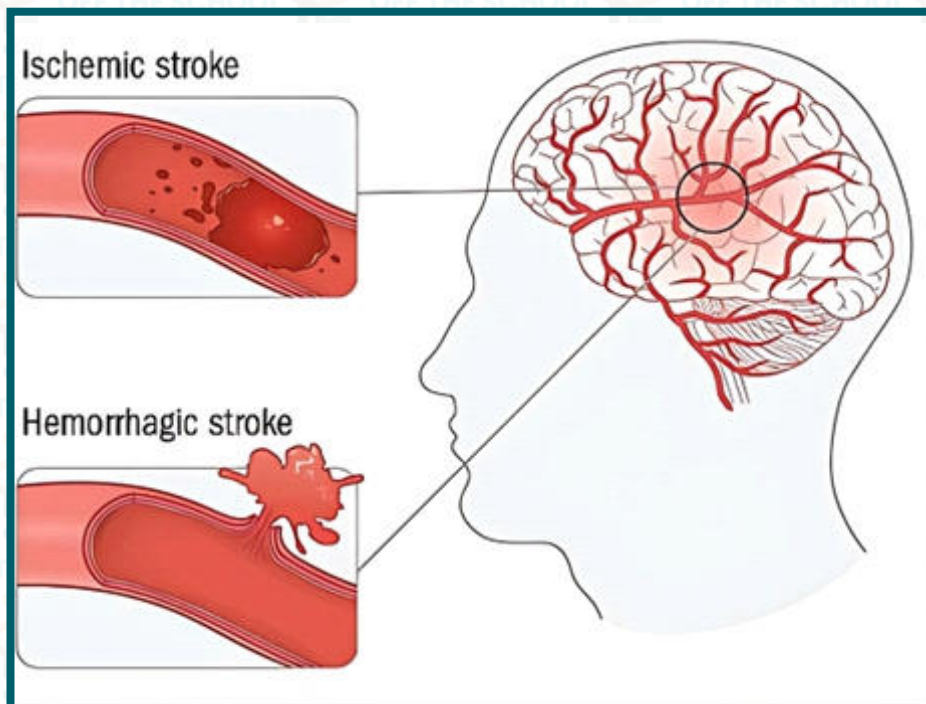
- **Smoking**
- **Diabetes mellitus**
- **High alcohol intake**
- **Cocaine abuse**
- **Hypertension** (high blood pressure)

Symptoms of Stroke:

- **Loss of balance and coordination**
- **Severe and sudden headache**
- **Paralysis** (typically affecting the arm, face, and leg)
- **Weakness** on one side of the body
- **Speech difficulties** and cognitive impairments may also occur depending on the area affected.

Treatment of Stroke:

- **Medical emergency:** Immediate treatment is essential to save lives and reduce long-term disability.
- **Restoring blood flow:** If the stroke is caused by a clot (ischemic stroke), treatments like **anticoagulants** and **platelet aggregation inhibitors** are used to break the clot.
- **Blood pressure management:** To prevent further complications and damage.
- **Nursing care:** Supportive care is necessary for recovery, including physical therapy for rehabilitation.



Hematoma:

- **Definition:** A hematoma is a **massive accumulation of blood** in the space between the brain and its outer covering (meninges).

Cause of Hematoma:

- **Hypertension:** High blood pressure is a major cause of hematomas.





Symptoms of Hematoma:

- **Loss of consciousness**
- **Sudden confusion**
- **Pale skin color**
- **Seizures**

Treatment of Hematoma:

- The treatment depends on the severity of the hematoma and the extent of the damage to the brain.
- Surgical intervention may be needed to drain the blood and remove any blood clots, especially if the hematoma is large or causing significant pressure on the brain.



Infectious Disorders of the CNS:

Meningitis:

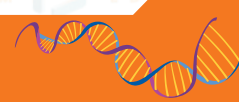
● **Definition:** Inflammation of the **meninges**, the membranes surrounding the brain and spinal cord.

Causes of Meningitis

- **Viral infections:** Most common cause.
- **Bacterial infections:** Second most common cause.
- **Fungal infections:** Rare, but possible.

Symptoms of Meningitis:

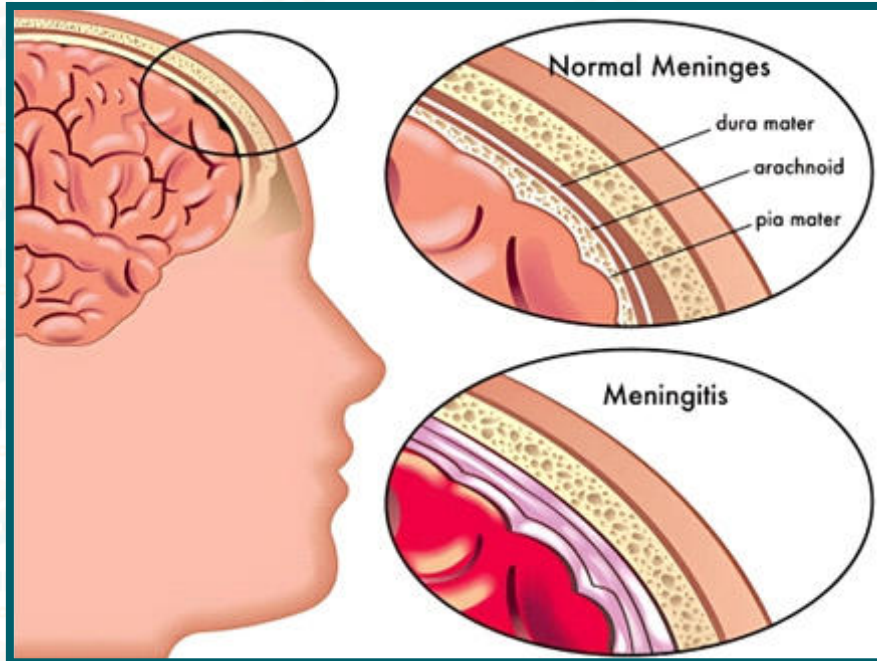
- **Sudden high fever**
- **Neck stiffness**
- **Severe headache**
- **Nausea or vomiting**
- May lead to:
 - **Paralysis**
 - **Coma**
 - **Death** if untreated.





Treatment of Meningitis:

- **Bacterial meningitis:** Requires immediate treatment with **intravenous antibiotics** and sometimes corticosteroids.
- **Viral meningitis:** Cannot be treated with antibiotics, but most cases improve on their own in a few weeks.
- **Transmission:** Spread through **airborne routes** such as **coughing** and **sneezing**.



Encephalitis:

- **Definition:** Inflammation of the **brain** itself.

Causes of Encephalitis:

- **Viral infections:** Most common cause.
- **Fungal infections:** Rare cause.

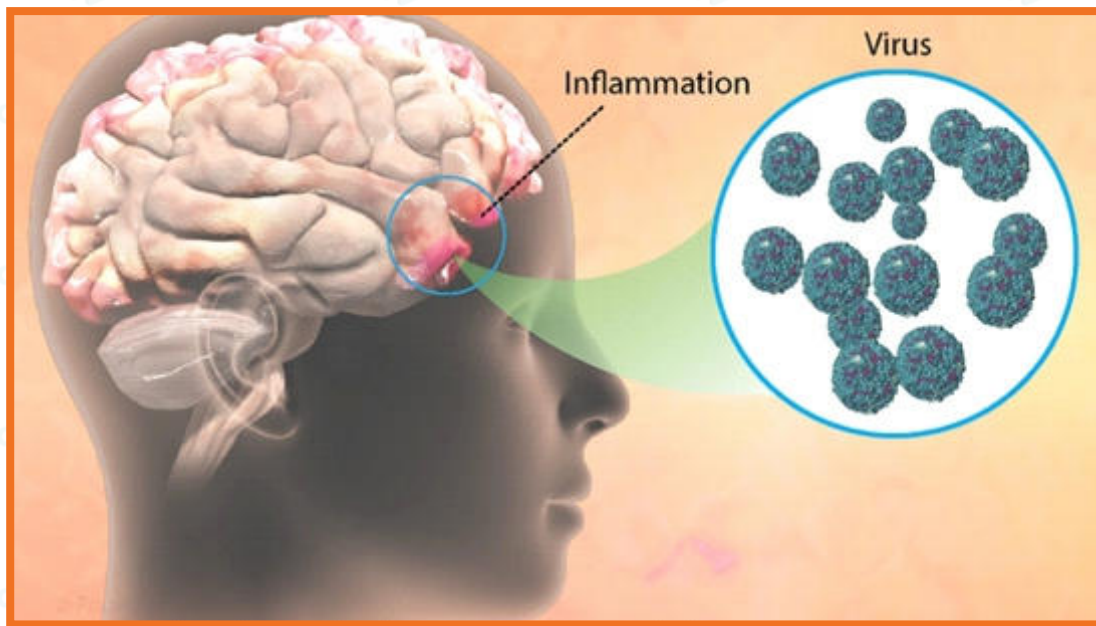
Symptoms of Encephalitis:

- **Fever**
- **Headache**
- **Confusion**
- Can lead to:
- **Muscle weakness**
- **Dementia**
- **Irritability**

Treatment of Encephalitis:

- **Urgent treatment** required.
- Treatment focuses on:
- **Addressing the underlying cause** (e.g., antiviral medications for viral encephalitis).
- **Symptom relief.**
- **Supporting bodily functions** while the body recovers.





Structural Disorders of the CNS:

Tumor:

- **Definition:** A **brain tumor** is an abnormal and uncontrollable growth of cells in the brain.

Causes of Tumor:

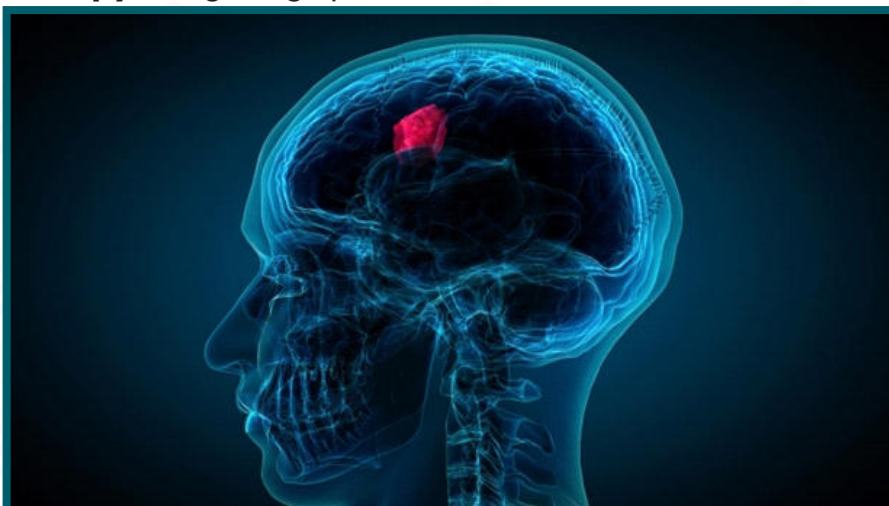
- **Mutation:** Genetic changes in cells that lead to uncontrolled growth.
- **Radiation exposure:** Can increase the risk of developing brain tumors.

Symptoms of Tumor:

- **Headache:** Often a common symptom, especially as the tumor grows.
- **Vision problems:** Blurred or double vision.
- **Seizures:** Uncontrolled electrical activity in the brain.
- **Paralysis:** Loss of function or weakness in part of the body.
- **Coma:** Loss of consciousness or awareness.
- **Death:** In severe cases, depending on the size and location of the tumor.

Treatment of Tumor:

- **Surgery:** To remove the tumor.
- **Radiation therapy:** To target and destroy tumor cells.
- **Chemotherapy:** To kill cancerous cells and reduce tumor growth.
- **Targeted therapy:** Targeting specific molecules involved in tumor growth.





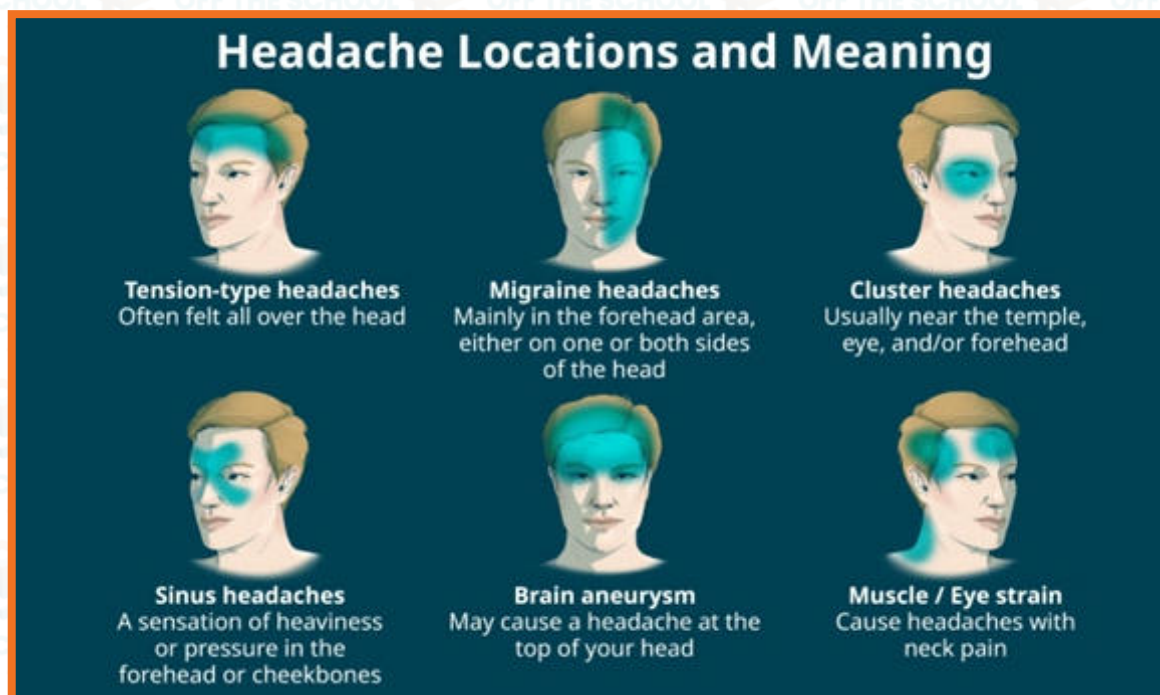
Functional Disorders of the CNS:

Headache:

● **Definition:** Pain originating from the head or upper neck, affecting the tissues and structures surrounding the skull. The brain itself doesn't have pain-sensing nerves.

Categories of Headache:

- **Primary Headaches:** These headaches are not caused by another underlying condition.
 - **Migraine:** Often accompanied by nausea, sensitivity to light, and sometimes aura (visual disturbances).
 - **Tension headache:** Caused by stress and tension in the muscles around the head and neck.
 - **Cluster headache:** Severe, recurring pain typically on one side of the head, often around the eye.
- **Secondary Headaches:** These result from another underlying condition.
 - **Infection:** Sinus infection, meningitis, or encephalitis can lead to headaches.
 - **Dental pain:** Infected teeth can cause pain referred to the head.
 - **Brain bleeding:** A brain hemorrhage or injury may cause headache.
 - **Cranial Neuralgias and Facial Pain:**
 - **Definition:** Inflammation of cranial nerves that carry sensory signals to and from the head and neck.
 - **Cause:** Inflammation or irritation of structures surrounding the brain.
 - **Symptoms:** Pain starting from the back of the head and upper neck.



Summary:

- **Brain tumors** are structural disorders characterized by abnormal cell growth and require urgent medical intervention.
- **Headaches** are common functional disorders of the CNS, classified into primary and secondary types based on the underlying cause.





Degenerative Disorders of the CNS:

Alzheimer's Disease:

● Definition: Alzheimer's disease is a progressive disorder where symptoms of dementia gradually worsen over time.

Cause:

● The exact cause is not fully understood, but it involves the degeneration of nerve cells in the brain, particularly in areas related to memory, thinking, and behavior.

Symptoms:

- Early Stage: Mild memory loss, difficulty recalling recent events.
- Mild Stage: Increased difficulty with memory and cognitive function; may experience delusions or hallucinations.
- Late Stage: Loss of the ability to carry on conversations and respond to the environment.

Treatment:

● No cure exists for Alzheimer's, but the goal of treatment is to manage symptoms and slow disease progression.



Multiple Sclerosis (MS):

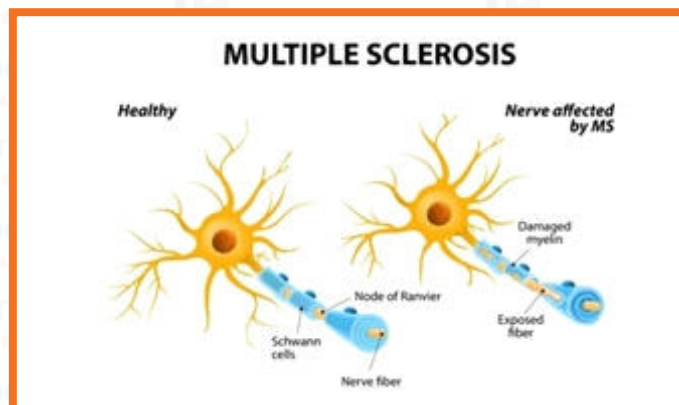
● Definition: MS is an **autoimmune disease** that affects the **central nervous system** (CNS).

Cause:

● The immune system attacks the protective covering of nerve fibers, causing **degeneration** and **scarring** of the nerves.

Symptoms:

● Symptoms vary but may include **muscle weakness**, **vision problems**, and **coordination difficulties**.





Huntington's Disease:

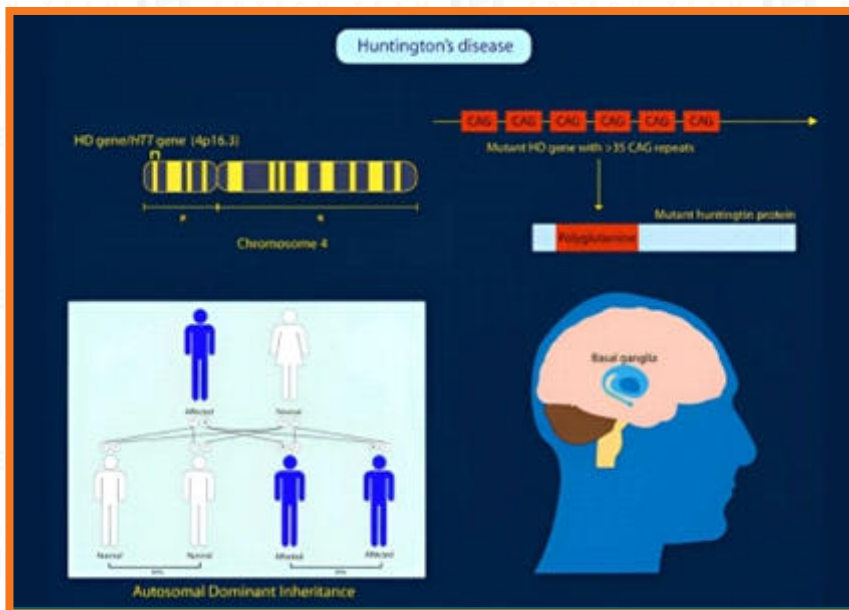
● **Definition:** Huntington's disease is a genetic disorder caused by the gradual breakdown of nerve cells in the brain.

Cause:

● The disease is **inherited** through a gene mutation, and symptoms typically develop in mid-adulthood.

Symptoms:

- Affects **physical movement, emotions, and cognitive abilities.**
- Early symptoms include **difficulties with movement** and **mood swings.**



Parkinson's Disease:

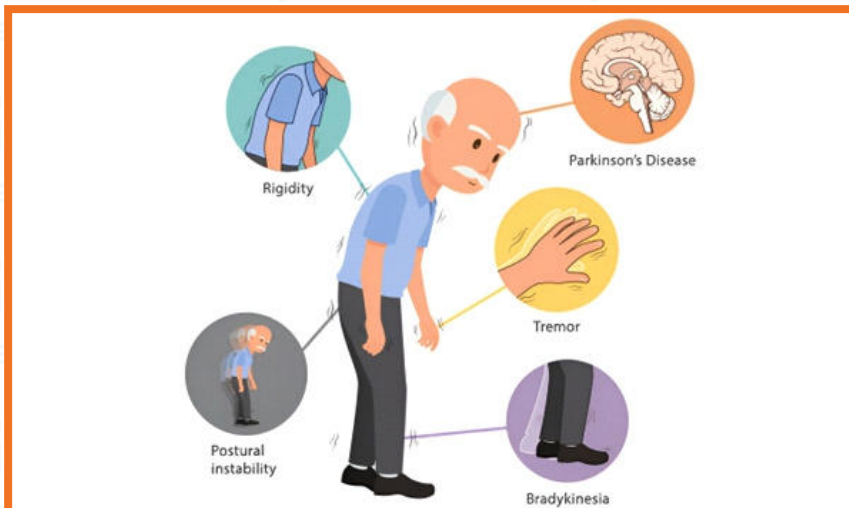
● **Definition:** Parkinson's disease is a **brain disorder** caused by **degeneration** or damage to **dopamine-producing neurons** within the **basal ganglia** of the brain.

Cause:

● The loss of dopamine-producing neurons leads to difficulties in **movement** and **coordination.**

Symptoms:

- **Tremors:** Particularly in the arm or leg, more noticeable at rest.
- **Muscle Stiffness:** Progresses to both sides of the body.
- **Weakness** and **trembling** of the muscles.
- **Difficulty with movement** and **balance.**





Diagnostic Tests for Nervous Disorders:

1. Electroencephalogram (EEG):

- **Definition:** An **EEG** is a neuroimaging test used to detect and record **minute changes in electrical activity** within the brain.
- **How it Works:**
 - It uses **macroelectrodes** (large, flat electrodes attached to the skin or scalp).
 - The test produces a **chart** (encephalogram) that shows how brain waves vary in terms of **frequency** and **amplitude** of electrical output from the brain over time.
- **Purpose:** It is used to monitor **brain activity** and detect disorders such as **epilepsy, sleep disorders**, and other brain abnormalities.



2. Computed Tomography (CT Scan):

- **Definition:** A **CT scan** is a **computerized X-ray imaging procedure** that produces detailed **cross-sectional images** of the body.
- **How it Works:**
 - The **X-ray beam rotates around the body**, and signals are processed by a computer to generate **tomographic images** (slices of the body).
 - These images provide more detailed information than conventional X-rays.
- **Purpose:** It is used to diagnose **brain injuries, stroke, tumors**, and other structural abnormalities in the brain.





3. Magnetic Resonance Imaging (MRI):

- **Definition:** MRI is a **radiological technique** that uses **magnetism, radio waves**, and **computer technology** to produce detailed images of body structures.
- **How it Works:**
 - The **MRI scanner** is a tube surrounded by a **circular magnet**.
 - The patient is placed on a **moveable bed** that is inserted into the magnet.
 - The magnet creates a strong magnetic field that aligns the **protons** in the hydrogen atoms of the body.
 - These protons are then exposed to a **beam of radio waves**, which makes them spin and produce faint signals. The signals are detected by the MRI receiver, and a **computer** processes this information to create an image.
- **Purpose:** MRI is widely used to obtain highly detailed images of the **brain, spinal cord**, and **nervous system structures** to diagnose disorders such as **brain tumors, multiple sclerosis**, and **nerve damage**.



Summary:

- **EEG** detects electrical brain activity and is useful for diagnosing conditions like epilepsy.
- **CT scans** provide cross-sectional X-ray images to evaluate structural issues in the brain.
- **MRI** uses magnetic fields and radio waves to create highly detailed images of body structures, particularly the brain and spinal cord, aiding in diagnosing neurological disorders.

